Optimizing Emergency Response with California's Environmental and Weather Factors

DAT 490 Data Science Capstone

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Abstract

Over the past decade, California has faced a drastic rise in the number and intensity of wildfires. This surge is attributed to the effects of climate change on the state's environment and weather patterns. From 2013 to 2023, an average of 8,000 wildfires occurred every year with each blaze burning around 1.3 million acres of land yearly (Frontline Wildfire Defense, 2020). In addition to the property destruction inflicted by the fires, the state suffers from catastrophic loss of life, displacement of communities, and considerable harm to biodiversity and the economy. The shifting land use practices, urban encroachment, and increased vegetation growth coupled with climate change has made wildfires more severe. Succeeding dry seasons have intensified temperatures and greatly decreased the rainfall, leading to the sudden change in weather patterns. These unforeseen instances of wildfires have raised concerns for emergency service providers and government bodies responsible for providing relief aid to the affected regions. This paper reviews the use of data science in devising efficient response methods for wildfires by analyzing the relationship between weather factors, environmental patterns, and wildfire occurrences and wildfire occurrences. Our goal is to provide insights for wildfire emergencies and decisions such as wildfire prevention, mitigation, and resource allocation. We employ machine learning, predictive modeling, and real time environmental data so California can identify new avenues to enhance the level of preparedness and efficiency in wildfire management to minimize the socio-economic and environmental consequences these disasters bring.

The challenging nature of managing wildfires stems from the erratic character of such events. The spike in the intensity of the wildfires often results in emergency services

being strained which when paired with poor coordination and delayed distribution of resources worsens the situation. An attempt to estimate wildfire activity based on traditional approaches and historical records is very common, but these methods fail to account for the dynamic nature of wildfires. This paper showcases a unique methodology that merges EDA and machine learning techniques for predicting wildfires and their severity. The scope of this research is to supplement historical wildfire data with numerous climate variables such as temperature change, precipitation, moisture in the soil, and wind speed to discern certain trends or correlations that will be helpful in predicting extreme wildfires. As far as the causes of wildfires are concerned, this research demonstrates the significance of environmental conditions, as well as the interaction between climatic factors and anthropogenic processes such as land use and city building. Through our modeling, we attempt to determine the potential danger of a wildfire in a certain area based on the current meteorological parameters, allowing true to life wildfires to be quelled and enabling emergency teams to have the best coordination. With the aid of high-risk area zone identification, resource allocation to vulnerable loci in advance, and fire spread pattern forecasting, the wildfire models of this study predict disastrous outcomes with reduced likelihood. Not only do we look to implement a user-friendly feature within the complex navigation array via interactive data visualizations as a phased approach, but also look to enhance the broader wildfire management features of the system. These visual tools increase coordination and help improve decision making by providing critical, real-time information on fire locations, weather, and evacuation routes. Through these means, we hope to be proactive

towards California's wildfires by making emergency response more efficient as well as reliable.

We follow a structured ethical framework while collecting and using data. In particular, the merger of large-scale environmental data as well as predictive modeling includes extremely sensitive personal information like home addresses, current evacuation statuses, and public health information which raises inequalities in data privacy and allocation. To the best of our ability and within the boundaries of respecting privacy and the rights of individuals, we prescribe obtaining maximum benefit from research and enforcement of stringent ethical norms on data collection, storage, and usage. This encompasses obtaining community consents where such information is collected, strict measures to anonymize sensitive information, and secure controlled information sharing. Moreover, we underscore the need for easier access to wildfire management resources particularly by low-income communities which suffer the most from such hazards. Our research argues that effective allocation of resources and designing evacuation strategies require higher levels of inclusive governance systems which consider the needs of the poorest and most vulnerable members of society. In addition, the use of data science in the manner described contributes towards greater overall sustainability and resilience in fire-prone areas, which in essence translates into reduced suffering and destruction of property during wildfires. Finally, this paper is an illustration of how the adoption of data science in wildfire management can improve the functionality and responsiveness of the emergency services.

By utilizing machine learning techniques with both past and current environmental data, danger specialists can more accurately predict when and where

wildfires are most likely to erupt, helping them leap into action to contain the fire. These methods allow for resources to be focused on the most at-risk areas, prevents escalation, and enables timely evacuations and containment efforts that saves lives and assets. The positive results of these techniques on California's wildfire management can also assist other regions that experience climate change and its consequences, showcasing how data science can vastly improve emergency management. In essence, this study highlights the importance of applying advanced analytical methods to environmental and weather data to improve wildfire response management, building stronger fortifications within and surrounding California, and potentially, the world.

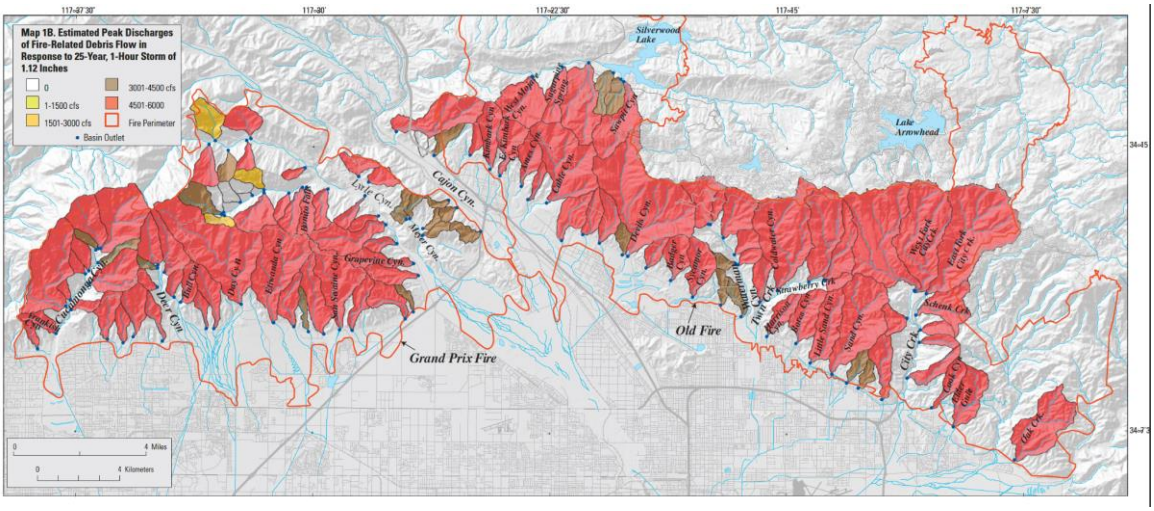
Project Plan

Introduction

Wildfires are inherently unpredictable and challenging to prepare for due to their dynamic nature. Alfred T. Westerling, a climate and fire scientist, highlights that the impacts of climate change will exacerbate this unpredictability. His research suggests that in the future, extreme wildfires, those exceeding 24,710 acres, may become 50 percent more frequent (as cited in National Academies of Sciences, Engineering, and Medicine). From 2013 to 2023, California averaged over 8,000 wildfires and 1.3 million acres of land burned every year (Frontline Wildfire Defense, 2020). A large number of resources are directed towards dealing with active wildfires and their lingering consequences. Understanding the drivers behind extreme wildfire events can help

enhance the direction of services for fire prevention, fire mitigation, and emergency response.

California Wildfire Overview

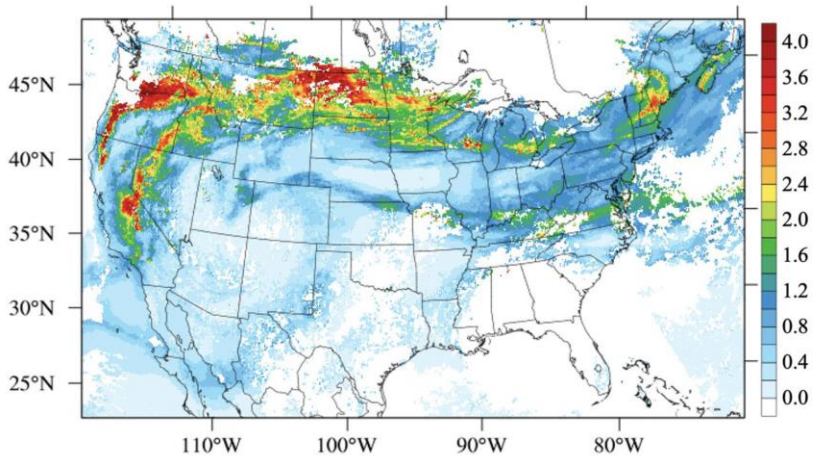


Estimated Peaks Discharges of Fire-Related Debris Flow (U.S. Department of the Interior & U.S. Geological Survey, 2005)

Estimated peaks of discharges of fire-related debris flow, one of the most common aftermaths of wildfires, is shown in the map above. Debris flow is dangerous due to the damages it causes to the environment, structures and human life. It is most likely to happen about 2 years after the fires, and “usually triggered by rainfall” (U.S. Department of the Interior & U.S. Geological Survey, 2005).

Many studies have shown that there is a strong correlation “between wildfire smoke exposure and exacerbations of asthma, chronic obstructive pulmonary disease, bronchitis and pneumonia”. The effects of wildfire smoke have a negative impact not on the environment but also on the quality of human life. According to the National Ambient

Air Quality Standards (NAAQS), several of the seven criteria air pollutants that have negative effects on the environment and human life are “emitted and formed by biomass burning” during wildfires. In the 2017 California wildfires, the wildfire smoke was estimated to have a 47% contribution to the mortality (Li et al., 2021, 2).



The fire smoke was transported across the continent to the eastern U. S. coast via the westerlies in the middle of September 2020 (Li et al., 2021, 2)

California Fire Statistics from 2018 to 2020

2018

Total Number of Wildfires (arson, smoking, camping, etc.) 3504

Total Acres Burned 1,063,414

Total Cost of Damages $4,084,573,016

2019

Total Number of Wildfires (arson, smoking, camping, etc.) 3086

Total Acres Burned 129,914

Total Cost of Damages $404,042,239

2020

Total Number of Wildfires (arson, smoking, camping, etc.) 3501

Total Acres Burned 1,458,881

Total Cost of Damages $4,264,918,165

Above details are the statistics of damages wildfires have on the environment and on human life (Statistics | CAL FIRE, n.d.). Motivated by the need to take action in preventing wildfires and reducing the damages, our team decided on researching and predicting the impact and severity of wildfires in order to assist authorities and local citizens in mitigating wildfires and enhancing emergency response.

Research Questions

Question 1

Can the level of wildfire risk (e.g. **Low, Moderate, High, Extreme**) for individual counties in California be predicted based on weather data to optimize emergency resource allocation?

Question 2

How does the population in urban areas impact correlated wildfire weather factors?

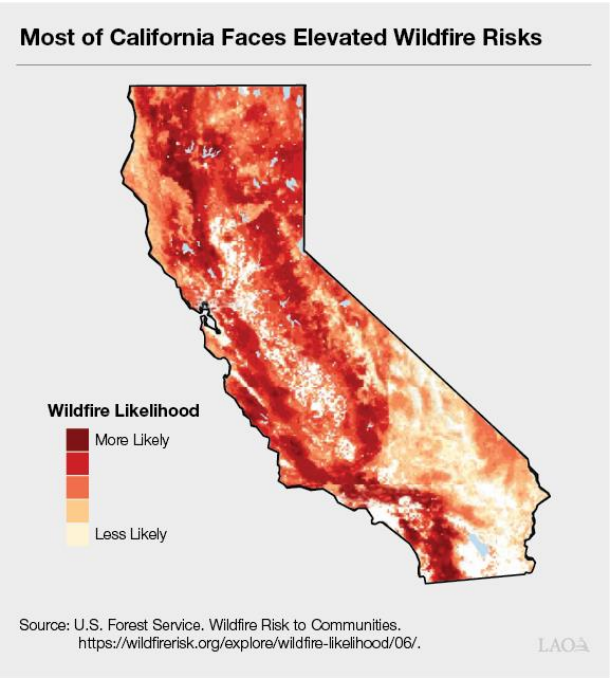
Hypotheses

Hypothesis 1: *The level of wildfire risk for individual counties in California can be predicted based on the weather data.*

Using Random Forest, K Nearest Neighbors, and Gradient Boosting models, the level of wildfires risk could be predicted based on the weather conditions of California.

Hypothesis 2: *Urban areas have an impact on weather factors that contribute to wildfires.*

By comparing the ability to predict the fire risk of each area (Urban, Rural, and Semi-urban), the conclusion whether urban areas have an impact on weather conditions that related to the wildfires could be made.



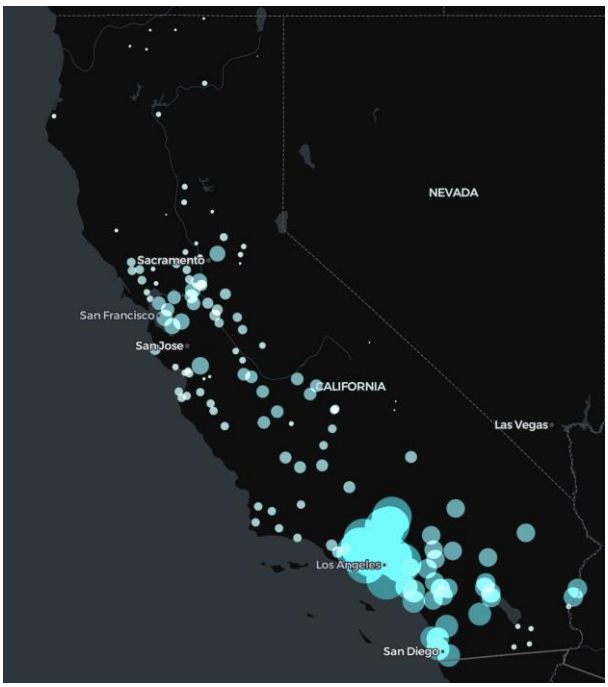


Image on the left presents the likelihood of wildfires in any given years (Legislative Analyst's Office, 2025), and the image on the right is generated by open

source geospatial visualization tool called Kepler, using main dataset (California weather conditions and county's location), and supporting dataset of California population by county (United States Census Bureau, n.d.). Based on the brief comparison between these two visualizations, it is believed that fires in urban areas will occur more frequently than in rural areas.

Data

Source

The main dataset (California Environmental Conditions) is originally gathered from the website of California Irrigation Management Information System (CIMIS), and organized by Chelsea Zaloumis on Kaggle. Several weather factors such as precipitation, temperature, humidity, wind speed, etc.) that contributed to the wildfires are listed in the dataset. Besides the main dataset, the geographical data of fire stations from California Open Data Portal is used to match the recorded regions to California's counties. In order to determine the severity of wildfires, a dataset of wildfires damages from California Department of Forestry and Fire Protection (CAL Fire) is also included.

Regions

Weather conditions were recorded daily across different CIMIS weather stations, which are regions in the main dataset.

Counties

Considering the research questions, additional variables such as counties, longitude, latitude, and elevation converted from supporting datasets are integrated with the main dataset to determine other factors such as wildfires severity and areas (urban, semi-urban, or rural) location.

Environmental Factors

To effectively predict the potential wildfires and their severity, multiple environmental factors are used in the main dataset. They include precipitation, solar radiation, vapor pressure, air temperature, humidity, wind speed, wind run, and soil temperature. Statistics are recorded daily for each station producing a total of 128,125 observations.

County’s Latitude and Longitude

The latitude and longitude of each county in California is also imported to support visualizing geospatial data.

Area of Each County

Each county in California is categorized into urban, rural, or semi-urban areas based on information from California State Association of Counties (California State Association of Counties, n.d.).

Computational Method and Output

For research question 1, comparison among the models will be conducted with key metrics including accuracy, precision, recall, and F1 score. Among these, the F1 score and recall are the primary metrics of focus, as the misclassification of high or extreme fire risk situations as low risk poses the greatest potential harm. Upon comparison of each model's performances, additional methods may also be explored. To help mitigate overfitting and gauge the degree of each model's generalization capability, K fold cross validation will be used.

Research question 2 investigates the impact of population to the weather conditions contributing to the wildfires. Specifically, urban areas (with high population density) will be the focus and be compared to the other areas. The goal is to find if there is a significant difference in the ability of predicting wildfire risk using weather conditions for three segments of data, associated with three areas. Machine learning methods like research question 1 will be used to predict the wildfire risk, and similar evaluation metrics such as precision, recall, and F-1 scores will be used to determine the performance of each model.

Output Summary

Question 1

Can the level of wildfire risk (e.g., Very Low, Low, Moderate, High, Very High, Extreme) for individual counties in California be predicted based on weather data to optimize emergency resource allocation?

The output for question 1 will generate a classification ranging from low to extreme class which will represent the potential severity of a fire for a given county during a given day. Several real-world fire examples will be presented to highlight the predictive capability of the model. Another potential output is a prediction of the number of structures that may be damaged or destroyed for a specific severity alert level. These predictions can be used to aid in decisions regarding the deployment of emergency resources, evacuation management, and the scale of the aid response.

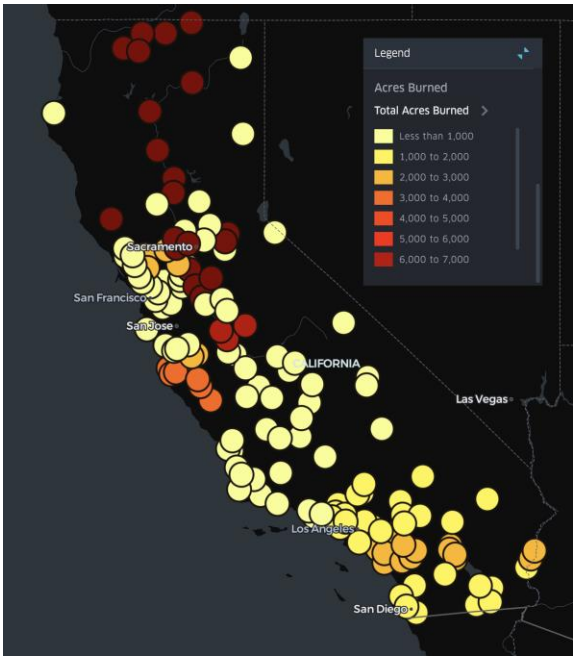
Question 2

How does the population in urban areas impact correlated wildfire weather factors?

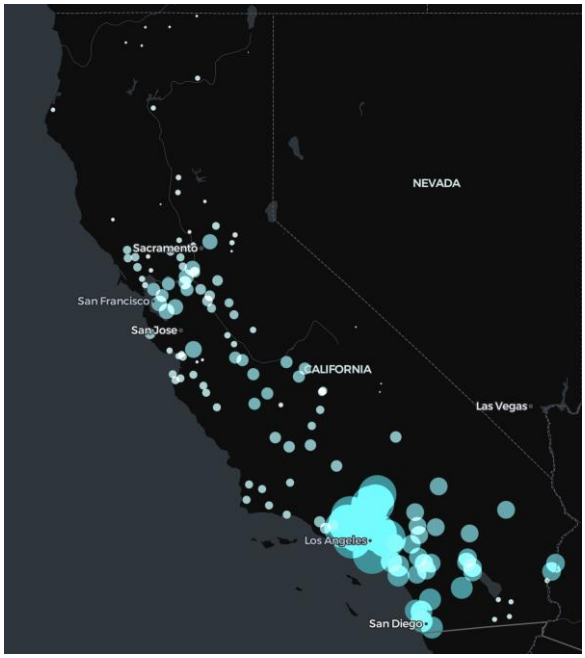
The main goal for this question is to determine if there is a significant difference in the ability of fire risk prediction among different areas reflecting different ranges of population. Each model's performance will be observed and analyzed. The improved model's performance of each area will be compared against each other to see if there is a significant difference.

Data Visualization

Since the main dataset contains time series data (weather conditions recorded daily from 2018 to 2020), visualizing how each weather factor changes over time is essential in further analyzing the weather pattern and addressing any possible abnormal data. Besides the main dataset, supporting datasets of California counties, locations, and wildfires damages contain geospatial data. Therefore, visualizing data associated with each county by mapping would also be a significant support to understanding the relationship between weather factors and geography factors. From there, further analysis could be conducted to improve the wildfire prevention systems and emergency response.



Total Acres Burned by Wildfires in Each County in 2018



Population Density associated with California counties in 2018

Main tools to support visualizing data would be any data visualization packages in **Python** (such as ggplot and seaborn), and **Kepler** - an open-source geospatial data visualization tool on the Internet. Above are examples of visualizing total acres damages based on counties and population density using **Kepler.**

Implementation

By identifying weather conditions that lead to more severe wildfires and implementing a model to predict their occurrence in question 1, preparing for the consequences like debris flows can begin earlier and directing services like fire prevention, evacuation planning, and aid response can all be enhanced. The model from question 1 is built on the two-year period from 2018 to 2020, but it can also be applied to predict present and future risks on a daily or even hourly basis. This index could help guide fire prevention efforts, such as aiding in the strategic and cost-effective deployment of drone surveillance in high-risk areas (Anderson 2023), implementing home defense measures like the collection of dry brush, and limiting the use of industrial equipment that could potentially spark fires.

Geographically, the analysis in question 2 can help identify which regions of California are more vulnerable to wildfire destruction. With this information, city and state municipalities can decide if the risks outweigh the benefits of building man-made structures and communities in those areas. Also, analysis of the frequency and severity of wildfires in urban versus rural counties can aid in future allocation of resources to improve services in underserved areas.

Exploratory Data Analysis

Since the goal of this project is to optimize emergency response with California's environmental and weather factors, it is important to quantify the severity of wildfires and the most important factors contributing to it. To assist in creating visualization and showing what our team is aiming for, additional information has been added to the original dataset, along with the merging of data from different sources into a separate smaller dataset. Additional variables added to the original dataset include:

Longitude, Latitude: To visualize the damage severity on a map, the geographic coordinates are imported to assist in locating the county on the map.

County: To serve the purpose of showing wildfires severity, the damage statistics and locations data were integrated. Counties are used in damage statistics data, while weather stations are used in the original data. Therefore, weather stations will be grouped to their respective counties.

Total Acres Burned Grouped by County

To have a better vision of where the destruction caused by wildfires takes effect, and how it has changed over 3 years (2018, 2019, and 2020), a satellite map using each county's coordinates, and the total acres burned categorized by year is generated. Using this map, we can also generate further analysis on the impact of surrounding nature on the intensity and severity of wildfires. Even though analyzing the surrounding

natural habitat is not our main goal in this project, this data remains crucial in understanding how other factors affect wildfires.

To effectively illustrate the total acres burned in each county, each weather station's coordinates will be assigned to its respective county. Some counties have more than 1 weather station, and a few counties have none. In addition, some counties are present in the original data but are not in the wildfire damages dataset. To successfully combine the geographic coordinates and damages, some counties have been grouped together. Counties that are grouped include:

Calaveras, Tuolumne, and Alpine → Alpine

Del Norte and Siskiyou→ Siskiyou

Glenn and Butte → Butte

Lake and Napa → Napa

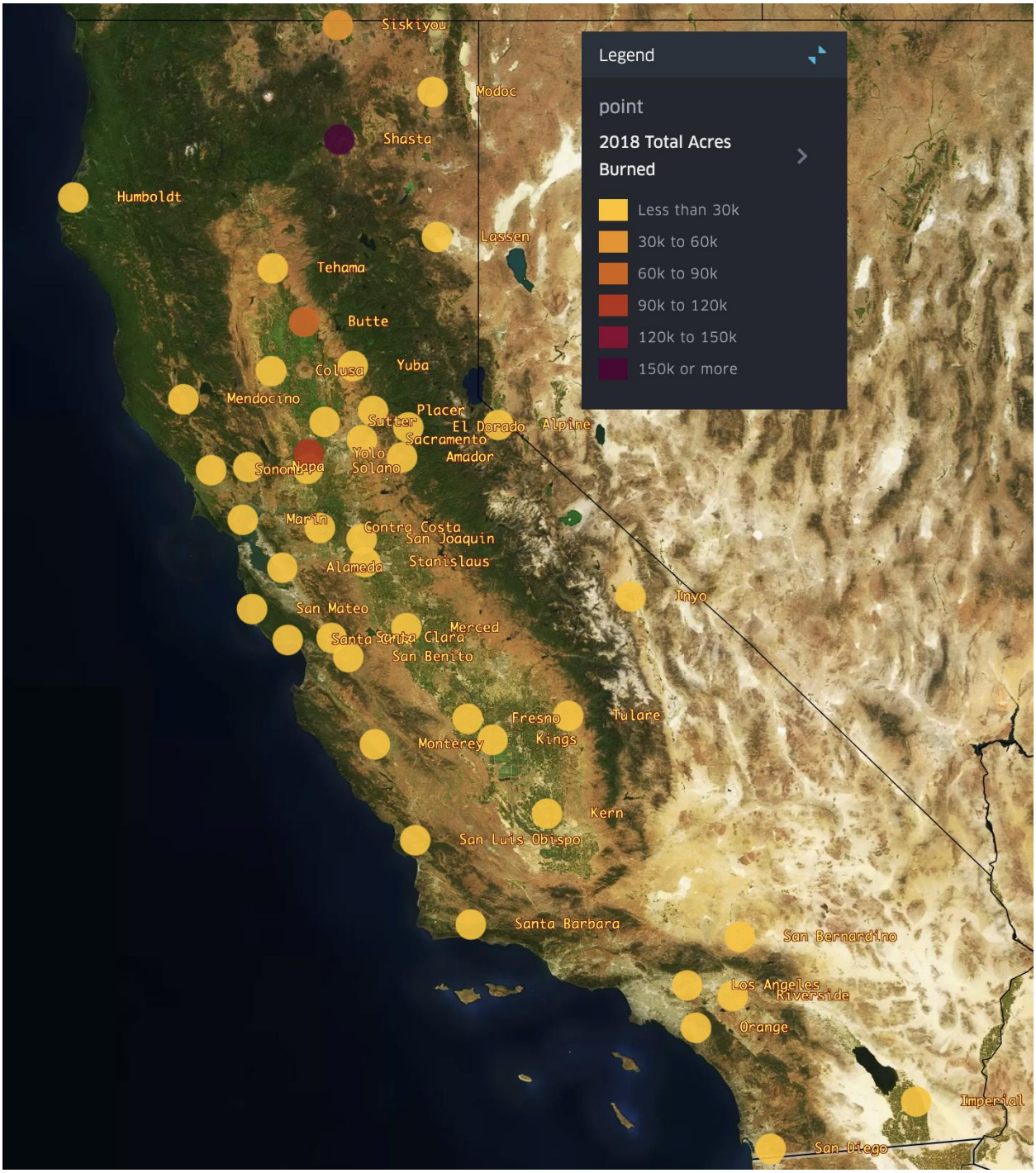
Madara, Mariposa, and Merced → Merced

Nevada and Placer→ Placer

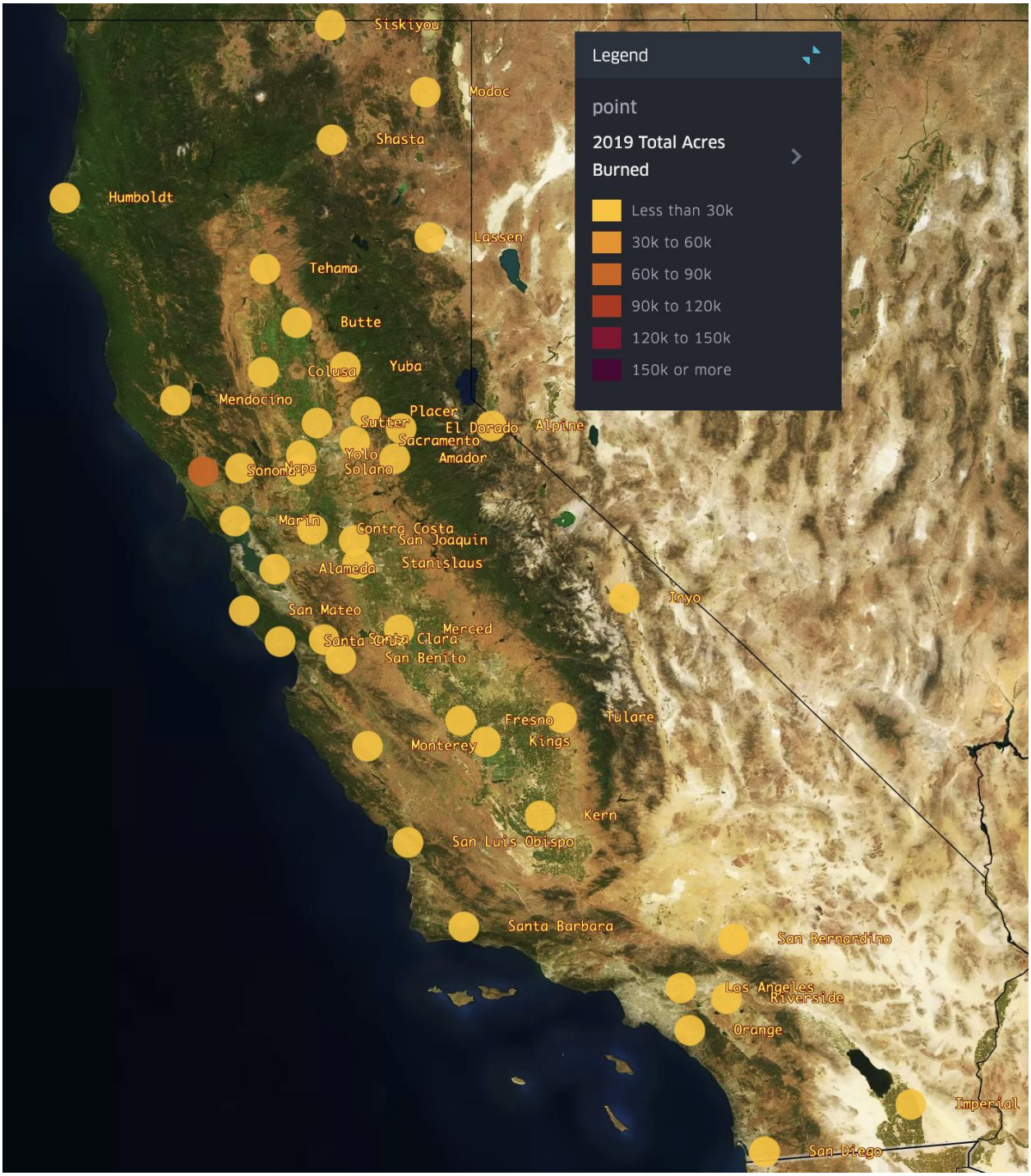
Trinity and Humboldt→ Humboldt

To avoid the influence of extreme values when grouping, grouped counties will share median damage values of all counties within the group.

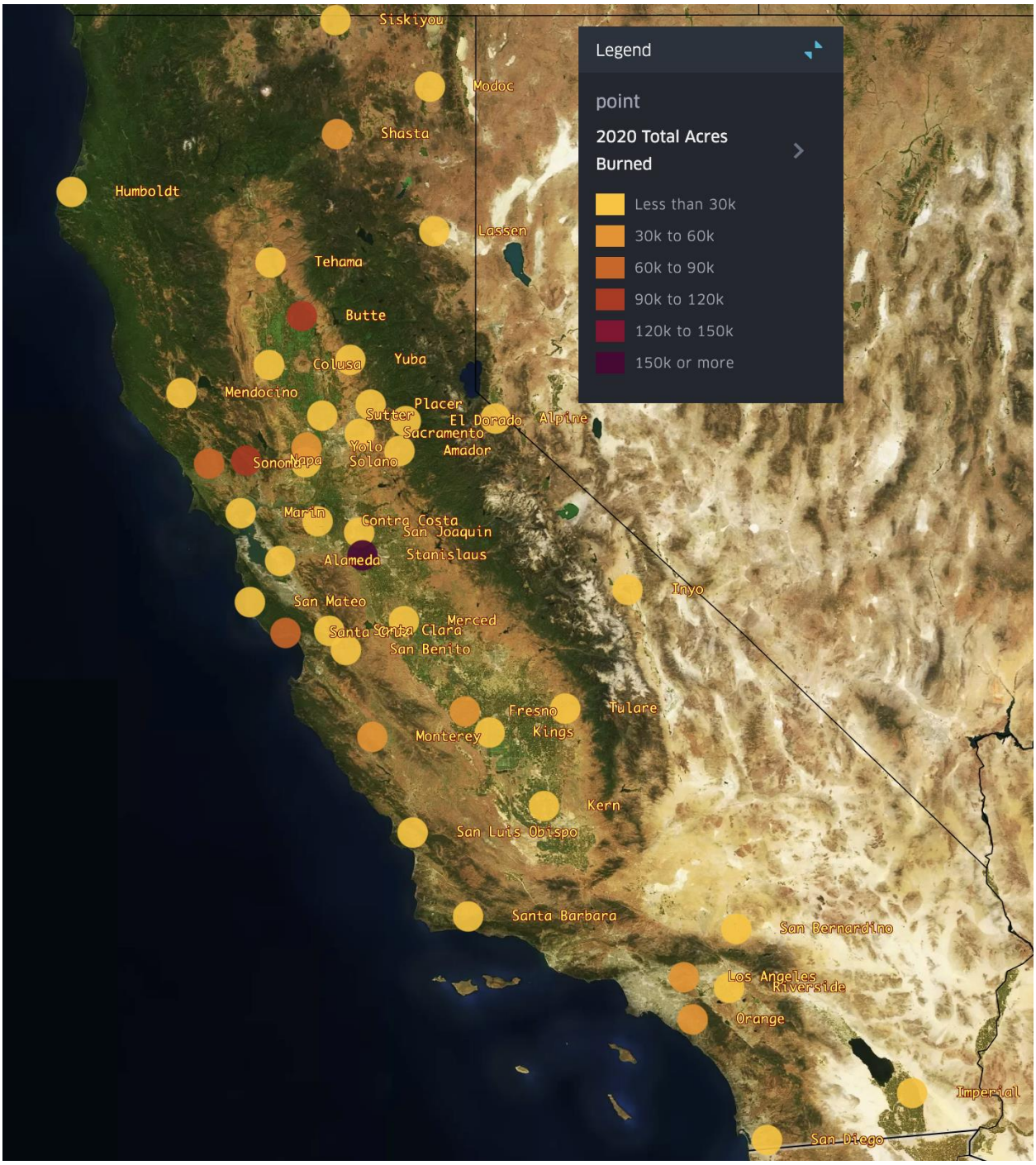
By showing the total acres burned associated with each county, we can have a better estimation of which counties in California need more attention than others in optimizing emergency response. This information will heavily impact the analysis for research questions 1 on the prediction of wildfire severity of each county based on weather factors.



2018 Total Acres Burned in Each California's County



2019 Total Acres Burned in Each California's County



2020 Total Acres Burned in Each California's County

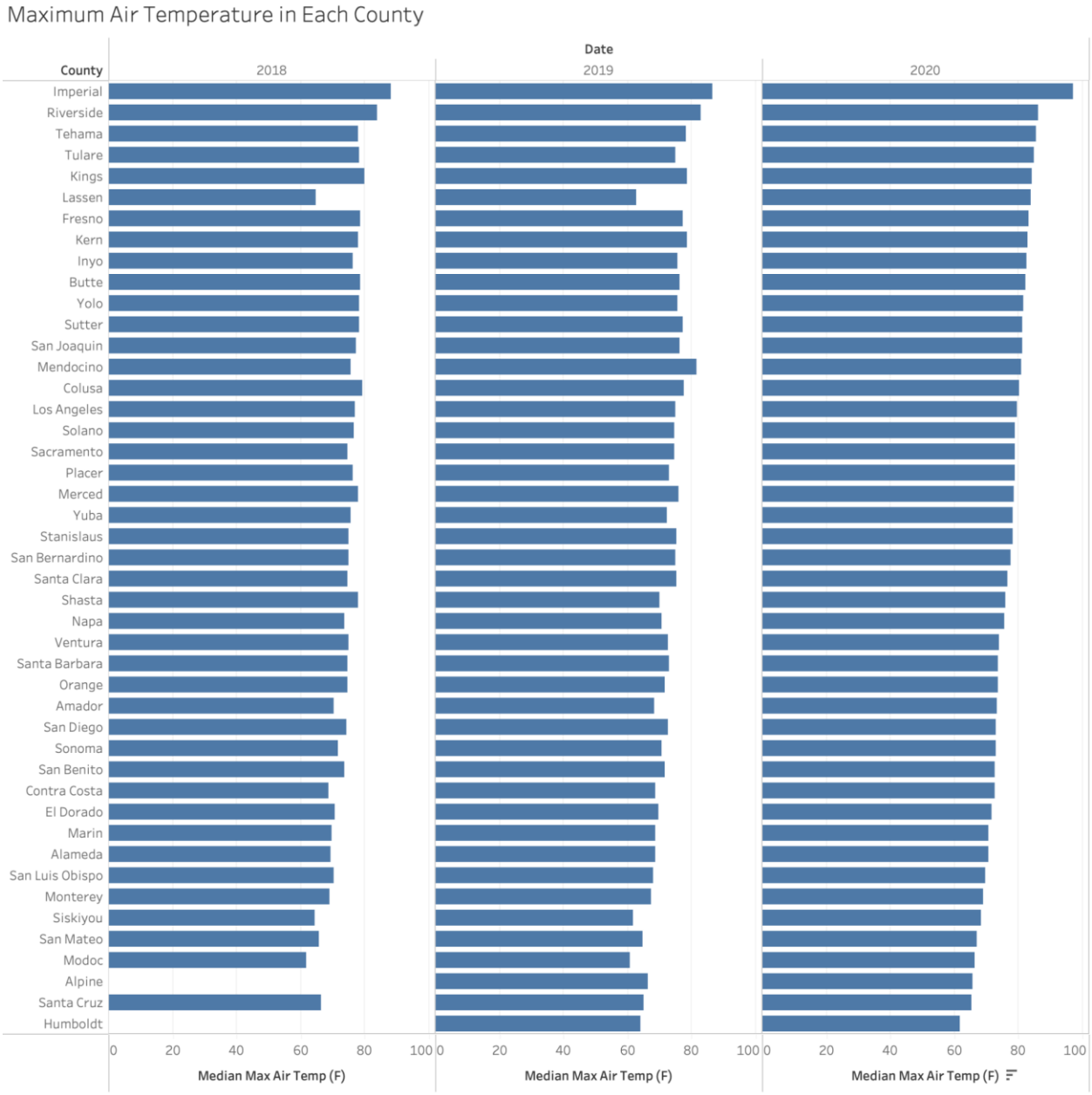
In the 2018 map, Siskiyou, Shasta, Butte, and Yolo are 4 counties that exceed 30,000 acres burned. The number of counties that have more than 30,000 acres reduced to one in 2019 - Sonoma. However, in 2020, the number of counties that lost

upwards of 30,000 acres almost tripled from 4 to 11. Not only did the number of counties with many acres burned increase, but the severity of fires also escalated in 2020 (see Fire Severity and Weather Factors graphs below). Based on this observation, deeper analysis and further research could be done to identify the reason why wildfire intensity increased significantly in 2020.

Maximum Air Temperature in Each California County from 2018 to 2020

Heat is always considered one of the most important factors contributing to fires. However, it might not be the most important one. When observing temperatures in each county individually, the top five counties that have the highest temperatures estimated are not the counties with a high number of acres burned. As illustrated in the below bar chart, it is visible that Imperial and Riverside counties have the two highest air temperatures recorded. However, there were 30,000 or fewer acres lost in 2018, 2019, and 2020 for those two counties. When viewed from satellites, it indicates that the landscape of Imperial and Riverside counties' high air temperatures did not cause extensive fire damage. The combination of limited vegetation, low diversity within plant life, and semi-urban areas greatly restricted the fuel sources for wildfires which

tempered the extent of the fire damage.



Median of Max Air Temp (F) for each County broken down by Date Year.

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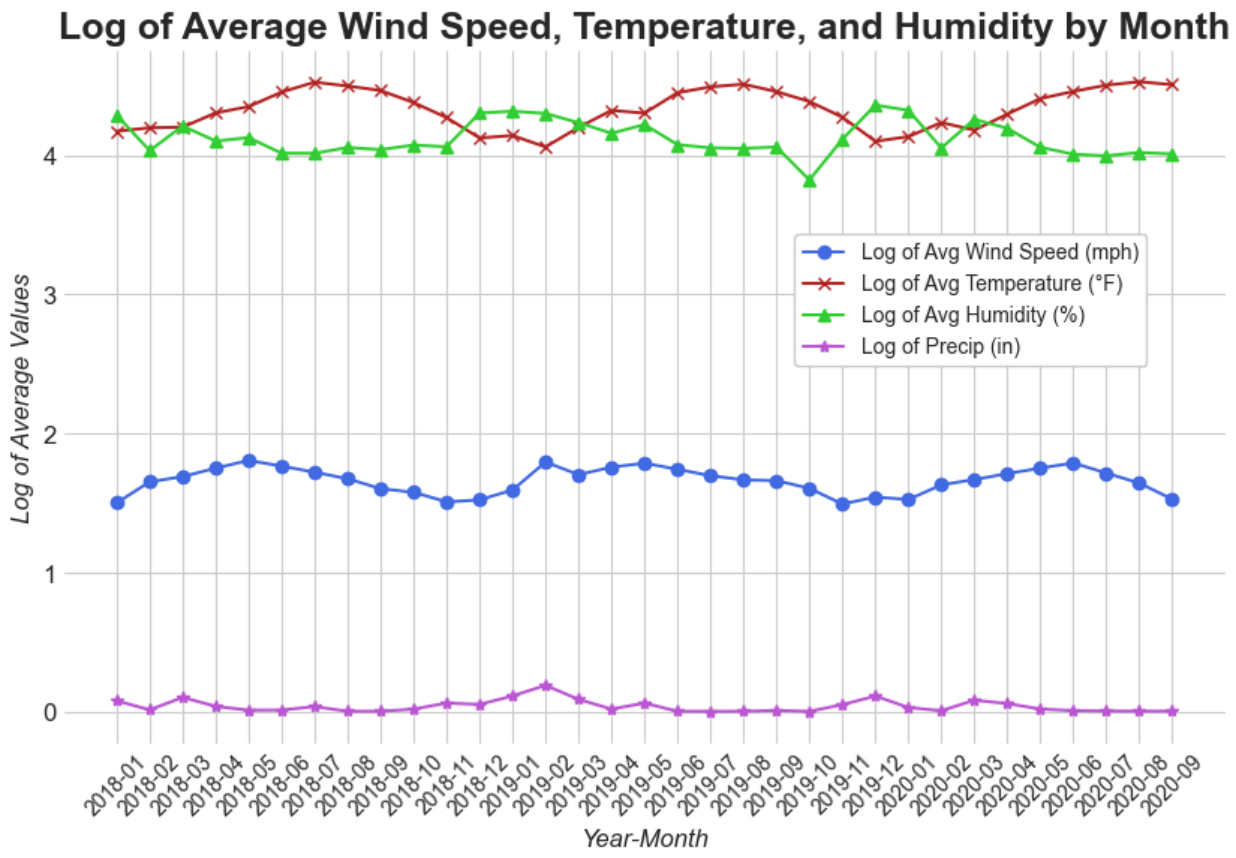
Weather Data

An examination of the uncleaned weather dataset reveals some issues that will need to be addressed before processing the data. Some days have null values for temperature and many CIMIS stations have dates missing due to damage or maintenance. Null values will be inputted using the median of the remaining valid measurements. Since missing dates may contain days on which fires occurred, the data on these dates will be derived using the mean of measurements taken on the same day in the same region. Each region resides in a unique climate, and this will ensure that the added days are consistent.

CIMIS Region	Number of Stations	Expected Total Days	Days with Records	Dates Missing	Days with Null Values
San Joaquin Valley	28	27776	24094	3682	13
Sacramento Valley	14	13888	11541	2347	14
Sierra Foothill	5	4960	4835	125	0
Bishop	4	3968	1727	2241	14
Imperial/Coachella Valley	12	11904	9327	2577	14
Northeast Plateau	8	7936	6705	1231	0
Los Angeles Basin	14	13888	12114	1774	7
San Francisco Bay	13	12896	12195	701	7
Central Coast Valleys	13	12896	9851	3045	21
South Coast Valleys	13	12896	12048	848	5
North Coast Valleys	8	7936	6742	1194	1
Monterey Bay	13	12896	11788	1108	7
San Bernardino	7	6944	4715	2229	13
Eastern Sierra Basin & Range	1	992	443	549	0
Totals	**153**	**151776**	**128125**	**23651**	**116**

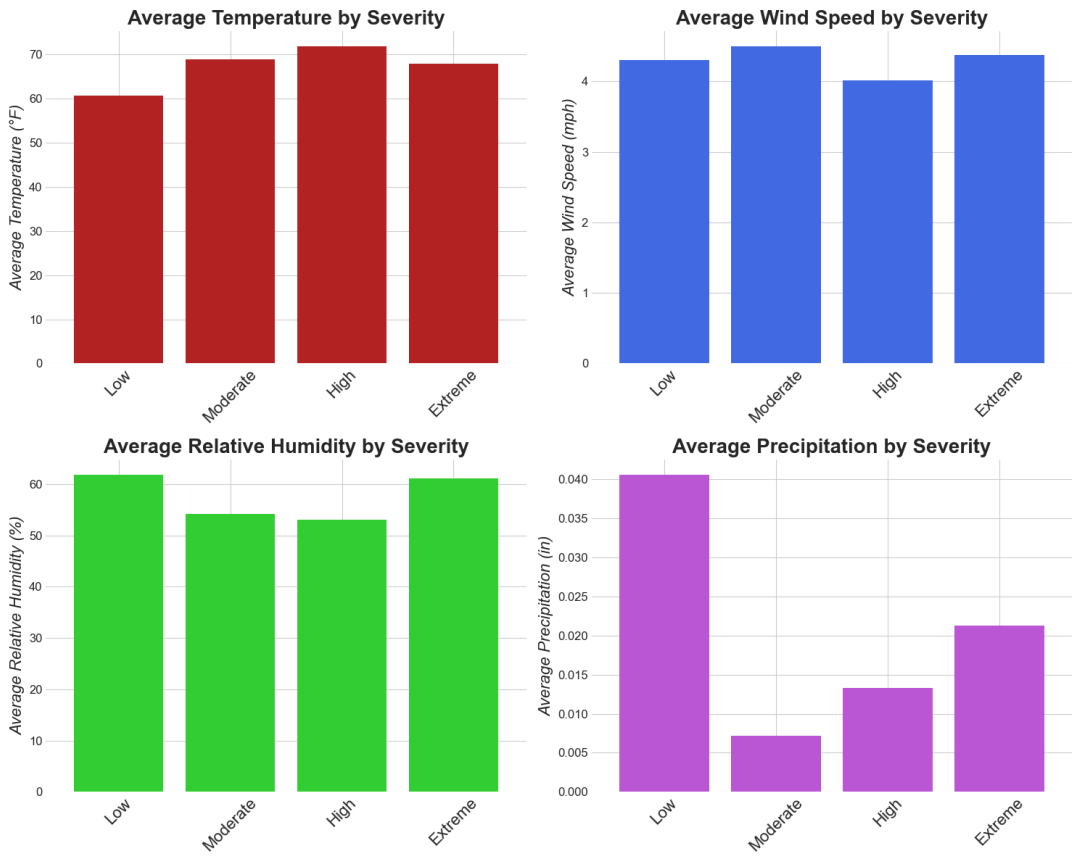
Weather Trends



A plot of key weather variables reveals some initial trends in the weather data. There is a clear seasonal trend with both temperature and wind speed, both usually peaking in the summer months. One notable exception is the wind speed in the winter months of 2019. Events like the Santa Ana winds, and the arrival of frequent storms can trigger high winds in the winter months, although it appears not to occur every year. The devastating Palisades fire which occurred during January 2025 was potentially exacerbated by a similar event. Humidity and precipitation tend to peak in the winter

months with seasonality as well. One exception is an early dip in the average humidity in early 2020 which may have helped to contribute to the larger fire count for that year.

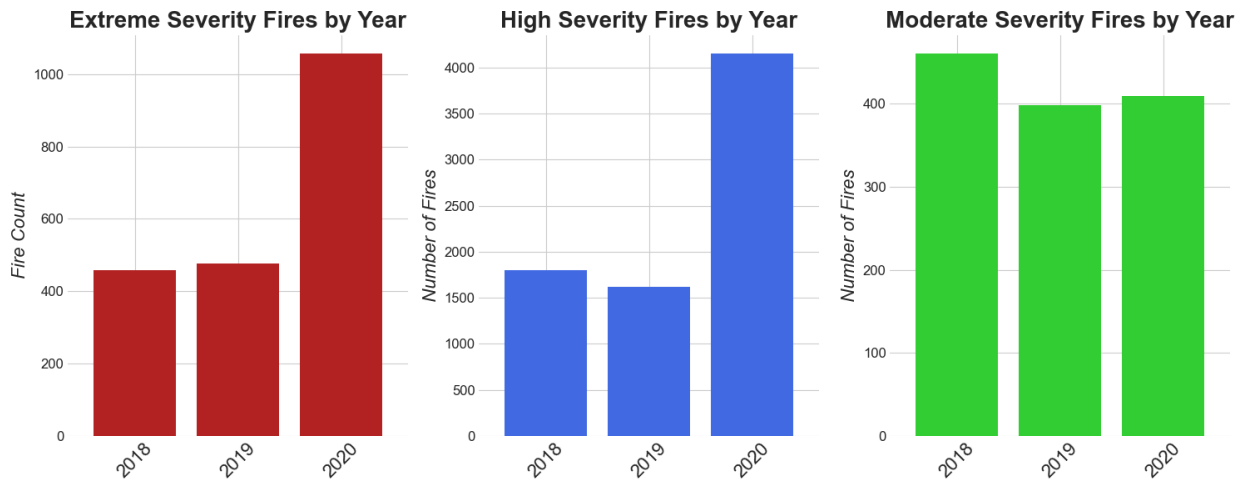
Fire Severity and Weather Factors



When broken down by category, some influences start to emerge. Immediately apparent is the difference in the amount of precipitation present with ‘Low’ risk (no fires reported) and days on which fires were present. These measurements appear to exhibit a strong initial correlation with fire severity, most likely due to the wet and dry seasons inherent to California’s climate. Precipitation appears to have some impact on the severity of a fire and a strong influence on the likelihood of ignition. The average temperature shows a clear trend, steadily increasing as fire severity also increases.

Wind speed, on its own, initially appears not to directly affect the severity of a fire. Data processing will include the creation of lagged variables to further explore these relationships.

Wildfire Severity Through the Years (2018-2020)



There were 278 fire events in the fire damage dataset. The most severe fires lasted for multiple days and some range over multiple counties. Since positive events are infrequent in the dataset, the damage from a single fire will be distributed across all CIMIS stations within the county and assigned to each day the fire remains active. Each county averages 7.7 days of fires burning per year.

An analysis of the count of severe fires by year shows a notable trend. In 2020, there is a significant increase in the number of high and extreme severity fires, while the frequency of moderate-severity fires remained consistent. Of the 278 total fires spread over the period of 2018 to 2020, 45.7% of them were directly caused by humans from

activities like vehicle use, campfires, industrial equipment use and arson. Many of these human caused fires could be prevented or mitigated by enacting safety measures during high and extreme risk environments.

Methodology

Research Question 1

Can the level of wildfire risk (e.g. **Low, Moderate, High, Extreme**) for individual counties in California be predicted based on weather data to optimize emergency resource allocation?

Classification machine learning methods are a powerful approach to incorporate historical weather information along with records of wildfire damage to generate a predictive model. Research uncovered two datasets to use for this purpose: the California Environmental Conditions dataset from 2018 to 2020, and the California Wildfire Damage Dataset. The environmental conditions dataset includes daily measurements of temperature, humidity, wind speed, precipitation, and solar radiation for numerous CIMIS stations located in the state of California. The wildfire damage dataset contains information about every fire that occurred in a similar time period including the number of structures damaged or destroyed, number of fatalities, and total acres burned. The datasets are merged with the county and date fields being used as common keys. The goal is to use the weather data from CIMIS stations to predict and classify wildfire severity risk into one of four levels: Low, Moderate, High, or Extreme.

With a categorical prediction of severity for the majority of counties for any given day, this index may help emergency response teams anticipate potential fire hazards and allocate resources effectively.

Step 1: Data Preprocessing

Preprocessing of data is essential to ensure the most accurate and competent results when modeling data with machine learning techniques. Formatting issues and missing values represent the initial hurdles in cleaning the dataset. Some numeric fields have formatting issues, such as those values containing commas and being classified as a string. These columns will be converted into either float or integer numerical formats. Null values underwent imputation and were replaced with the median value of all the other readings for the day. To deal with stations that contain missing dates, these dates were imputed using the mean values of readings from other stations in the same region and on the same day. Some stations had very few days with recorded readings, these are dropped from the dataset. To make the fields easier to work with and avoid problems with merging data, spacing, and case will be standardized throughout all columns of both datasets. Once the data is cleaned, MinMax scaling is applied to all numeric columns.

To create a measurable level of severity for each fire, key original factors from the dataset will be used along with additional engineered features. Engineered features will include:

- **Total Structural Damage**: Calculated by adding the number of destroyed and damaged structures.

- **Total Fatalities**: Derived by the sum of firefighter and civilian casualties.
- **Fire Severity Index**

The fire severity index was engineered by applying weights to each relevant indicator (20 % for **total acres burned,** 30 % for **total structural damage,** and 50 % for **total fatalities**), and then combined with multiplication. This numerical index was then categorized into ranks (Low, Moderate, High, Extreme) by quantiles.

To help test the theory that wildfire severity is influenced by both past and present weather conditions, lag features will be created for specific weather variables to represent trends. These include the average temperature, precipitation, humidity, and wind speed over the previous 7 days.

Given that the weather data contains 922 days of climate measurements for each county and the total number of fires in that period across all counties is 278, it is apparent that counties with a fire present are a rare occurrence. This may cause issues related to class imbalance and may present a significant challenge with most methods. During preprocessing, severity categorical rankings were spread among each day that a fire burned. This creates more positive values in the dataset and considers the spread of a fire. In addition, class balancing was tested in three ways: with built-in mechanisms, implementing Synthetic Minority Oversampling Technique (SMOTE) to artificially raise the number of minority cases, and Random Under Sampling (RUS) to reduce the majority class.

Step 2: Factor Analysis

Initially, relationships among factors with the wildfire severity index will be studied before deciding on the variables to be selected for modeling. A **correlation heatmap** generates the Pearson correlation coefficient for each factor and will serve as the first approximate gauge in determining whether there is a relationship between weather variables and wildfire severity. This will also help check for dependency among the independent variables. Factors that exhibit collinearity may be either combined or one will be dropped and those that exhibit no relationship may also be removed from the model.

Feature Interactions

In order to account for fire behavior on more complex environmental contributions to fire severity, new terms will be generated by multiplying two or more weather factors. These combinations are intended to focus on the particular conditions that may exacerbate fires.

Proposed interaction terms and their potential effects are:

- **Temperature × Wind Speed** - can result in Moderately fast fire spread.
- **Temperature × Precipitation** - can indicate the level of drought conditions.
- **Temperature × Precipitation × Wind Speed** - may be an indicator of having extreme conditions for fire.

These interaction terms will be examined for significance and if found, be incorporated into the modeling process along with others that strongly correlate with the

wildfire severity index. Correlating combinations may be utilized when the model is being calibrated to help refine the model's capabilities.

Step 3: Analysis

The three machine learning approaches that will be examined for multiclass classification are Random Forest, K-Nearest Neighbors (KNN), and Gradient Boosting (XGBoost). These models have different capabilities with respect to wildfire severity prediction. Their performances in classifying different levels of fire risk will be analyzed.

Random Forest Classifier: This technique utilizes multiple decision trees and can handle larger amounts of data of a more complex nature. It includes built-in strategies to combat overfitting, can accept categorical data, and has built-in feature importance scores that can help give insight into the relative importance of specific environmental factors. Although this method is expected to perform well, it adds more complexity to the model making it potentially more difficult to explain the final predictions.

K Nearest Neighbors (KNN): This algorithm classifies wildfire severity using the new cases that most closely resemble past cases. KNN is simple and easy to use, as it uses a voting process to assign a severity level based on the best-correlated neighbors. It is important to know that for KNN to be effective, an optimal value must be chosen for k. It is important to note that high-dimensional datasets, such as the weather dataset, can pose issues when distinguishing patterns and may cause the KNN model to perform poorly.

Gradient Boosting (XGBoost): This model may offer better results due to its improved handling of large data and its ensemble model approach. Unlike the previous approaches, gradient boosting excels at correcting for weaknesses as it proceeds by utilizing the gradient descent algorithm to combine multiple trees. Like the random forest classifier, it can provide feature importance scores to provide more insight into relationships. Tuned properly, this method has the potential to deliver the best outcome. However, it may require more complex hyper-tuning than the others for optimal performance.

Validation and Hyper-Tuning: Finally, a 70-30 train-test split will be applied to widely ensure that our models are performing well. Each model's performance will be analyzed using k-fold cross-validation to verify and avoid the issue of overfitting. Grid search optimization will also be employed for hyperparameter tuning with parameters including but not limited to tree depth and the number of estimators for Random Forest, the k-value for KNN, and the learning rate for XGBoost. The improved model will be tested on test data and the results will be compared and evaluated for the best-performing model in predicting wildfire severity.

Step 4: Evaluation Metrics

For the evaluation and comparison of each model’s performance, key metrics from a confusion matrix will be used including accuracy, precision, recall, and F1-score. Among these, the **F1-score**, and **recall** are the primary metrics of focus, as the misclassification of high or extreme fire risk situations as low risk poses the greatest

potential harm. Upon comparison of each model's performance, additional models or an ensemble approach may also be explored.

Step 5: Result Presentation and Visualization

This project aims to use weather data to generate a prediction of fire severity for a given CIMIS station location on any given day, past or present. To highlight this capability, example predictions will be generated and compared to real-world outcomes. From selected optimally performing models, feature trends and predictions will be presented and analyzed with visualization techniques. Correlation heat maps will be generated to highlight relationships among various features and geospatial visualizations, using tools like Kepler or Matplotlib, to illustrate the fire severity risk among counties in California. Finally, the project will consider strategies and policies that may be implemented to improve emergency response using a timely, accurate, and localized severity fire index.

Tools

For the analysis, extensive use of Python and its libraries, including pandas, numpy, sklearn, xgboost, geopandas, imblearn, seaborn, and matplotlib, will be used.

Research Question 2

How does the population in urban areas impact correlated wildfire weather factors?

By examining this, California local authorities could have appropriate plans (urban development on surface materials, building density, wildfire awareness activities, etc.) to help alter the aspect of weather that gets affected by urbanization.

Step 1: Data Preprocessing

The same preprocessing techniques applied in Research Question 1 will be utilized for Research Question 2 to ensure consistent formatting and exclusion of null values. Engineered Features such as Total Structural Damage, Total Fatalities, and Fire Severity Index will be carried over as well. From there the data will be further transformed by separating counties into 3 classifications: urban, semi-urban, and rural. Classifications will be assigned based on information provided by the California States Association of Counties.

Step 2: Analysis

For the analysis, the three machine-learning approaches, Random Forest, K-Nearest Neighbors, and Gradient Boosting, will be used for each classification group of urban, semi-urban, and rural. Once models have been created for urban, semi-urban, and rural populations, the same evaluation metrics utilized in Research Question 1 will then be applied (precision, F-1, Recall, and Support). Techniques such as k-fold cross-validation and grid search optimization, will be implemented to tune our data and avoid overfitting. The goal is to identify which model is the most accurate and the best predictor for each group within our classifications. This is different compared to question

one, which does not distinguish between classes but instead attempts to find the best overall predictive model.

Step 3: Evaluation Metrics

For the evaluation and comparison of each model's performance, the same metrics used in research question 1 would be applied: accuracy, precision, recall, and F1-score. Among these, the **F1-score**, and **recall** are still our primary metrics of focus, as the misclassification of high or extreme fire risk situations as low risk poses the greatest potential harm. For the urban area, the evaluation metrics of each method will be compared to identify the best one (using highest F-1 and recall score). This process will be repeated for semi-urban and rural areas. Finally, the evaluation metrics of the best method from each class will be compared against each other to see if there is a significant difference.

- If the evaluation metrics of the urban class are significantly different than the other classes:
    - If it is lower than the other classes, it implies that weather conditions in the urban area are not efficient in predicting fire severity. This means that the weather conditions in this class might be impacted by other factors in the urban area such as structures, building materials, etc., which makes it more difficult to predict the fire.
    - If it is significantly higher than the other classes, then it implies that the urban areas are impacting the weather conditions in a way that the

weather conditions could be easily used to predict the severity of fire (it could be because of less variety of topography, vegetation, etc.).

- If the evaluation metrics of the urban class are not significantly different than the other classes, then the weather factors are most likely the same across different classes, and that the weather factors are not impacted by the urban areas.

Data Visualization and Analysis

Research Question 1: Can the level of wildfire risk (e.g. **Low, Moderate, High, Extreme**) for individual counties in California be predicted based on weather data to optimize emergency resource allocation?

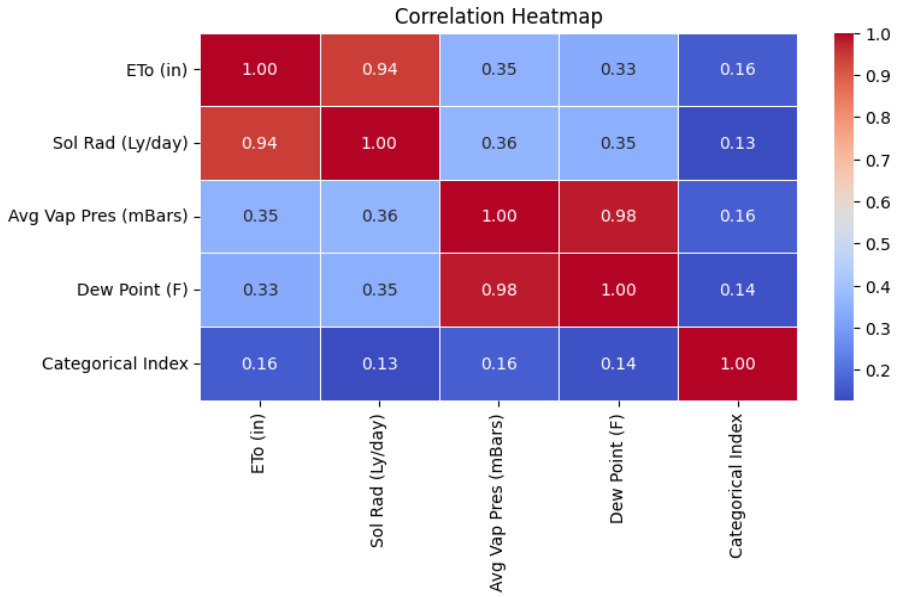
Variable Selection

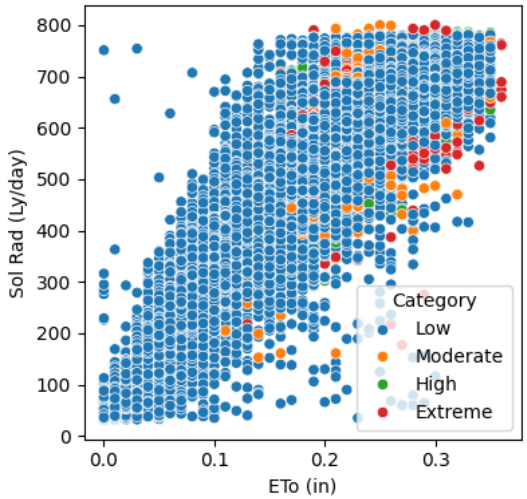
Variable evaluation and selection are key to enhancing predictive capacity in the models. Each potential factor was analyzed for correlation with the target (categorical fire severity) and for interactions due to collinearity. Lagged and combination features were also engineered and analyzed for interactivity.

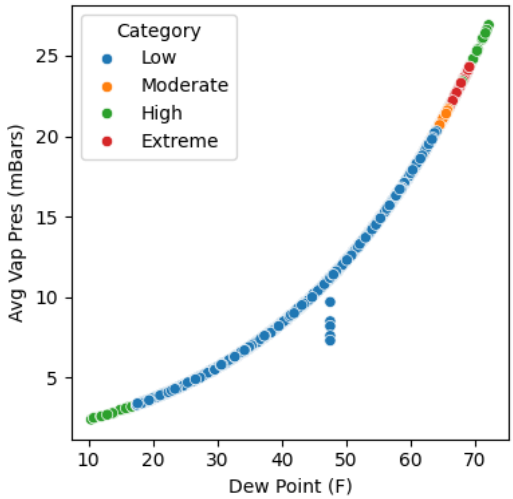
Removed Features

Eto (in) and Sol Rad (Ly/Day) exhibit a high level of correlation with each other. Solar radiation is a measure of the amount of sunlight in a day and ETo is a measure of evaporation. Avg Vap Pres (mBars) and Dew Point (F) also exhibit a high level of correlation. Interpretation of the interaction plots of these two pairs shows a clear dependency between the dew point and vapor pressure, while the relationship between the ETo and solar radiation is not as well defined. Opting to keep a measure of water

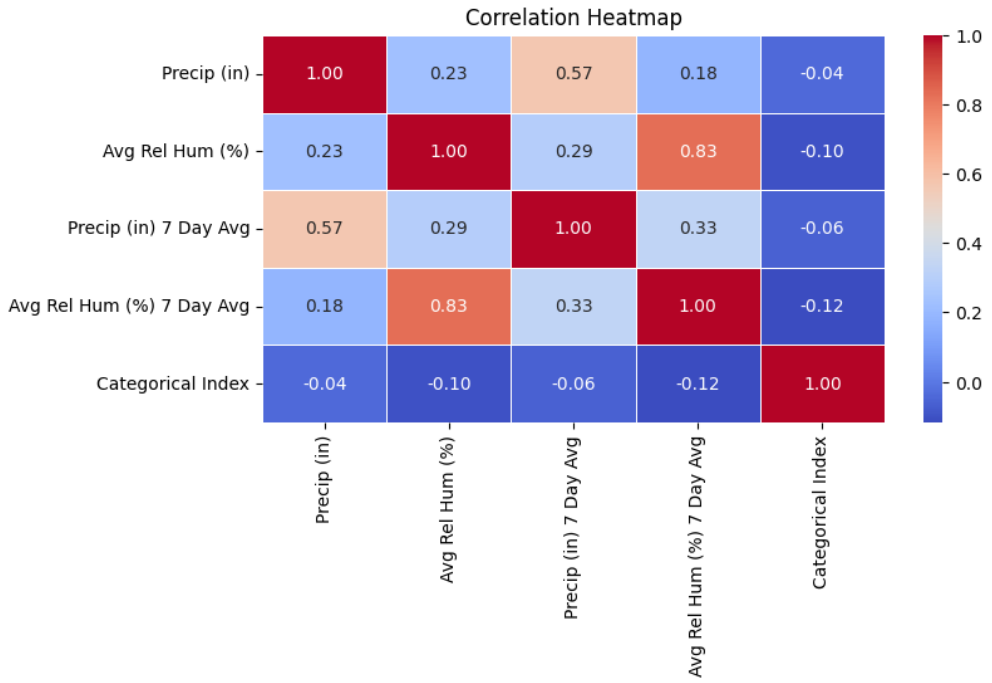
loss in the model, ETo is kept while solar radiation is dropped from analysis. Dew point is also removed from analysis in favor of vapor pressure.



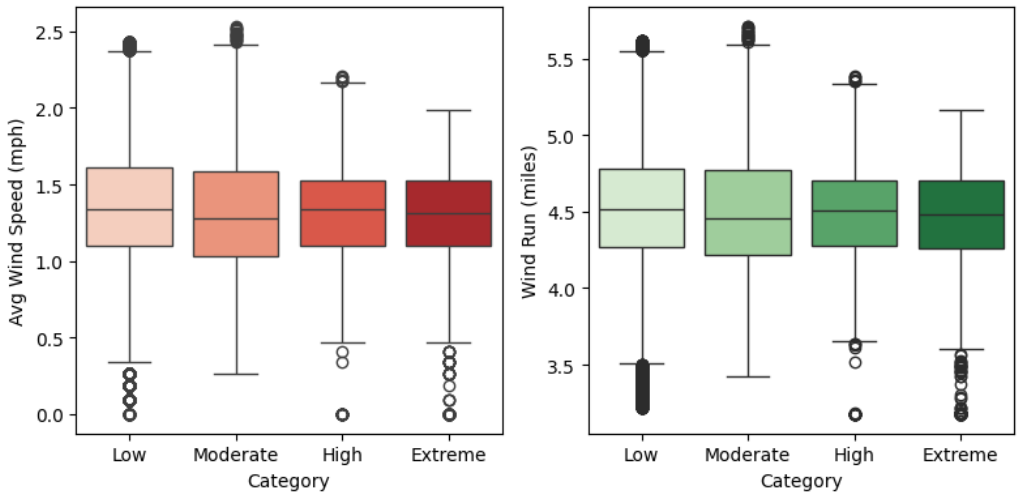




The moisture indicators involved in water supply, precipitation and humidity, show a varying degree of negative correlation with the severity of a fire. Although none of the features possess a strong correlation, to keep representation of water supply in the model, we opted to keep the average relative humidity over the past 7 days due to the slightly stronger relationship.



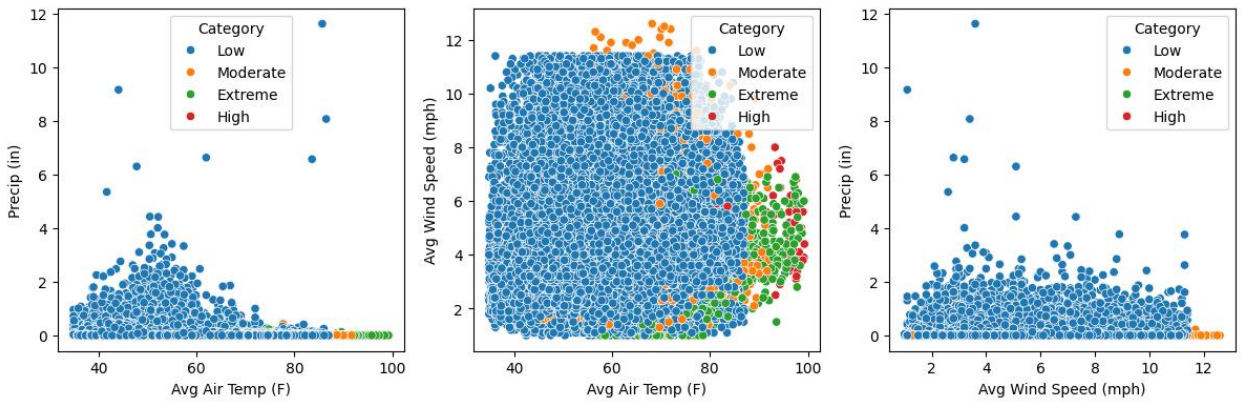
Both Avg Wind Speed (mph) and Wind Run (miles) show weak correlation with fire severity. Wind speeds are consistent overall in all severity conditions. Without combination with other relevant data, the lack of consistent correlation caused us to decide to remove both variables.



The minimum and maximum values for temperature and humidity are removed in favor of the average values as they all exhibit similar correlations.

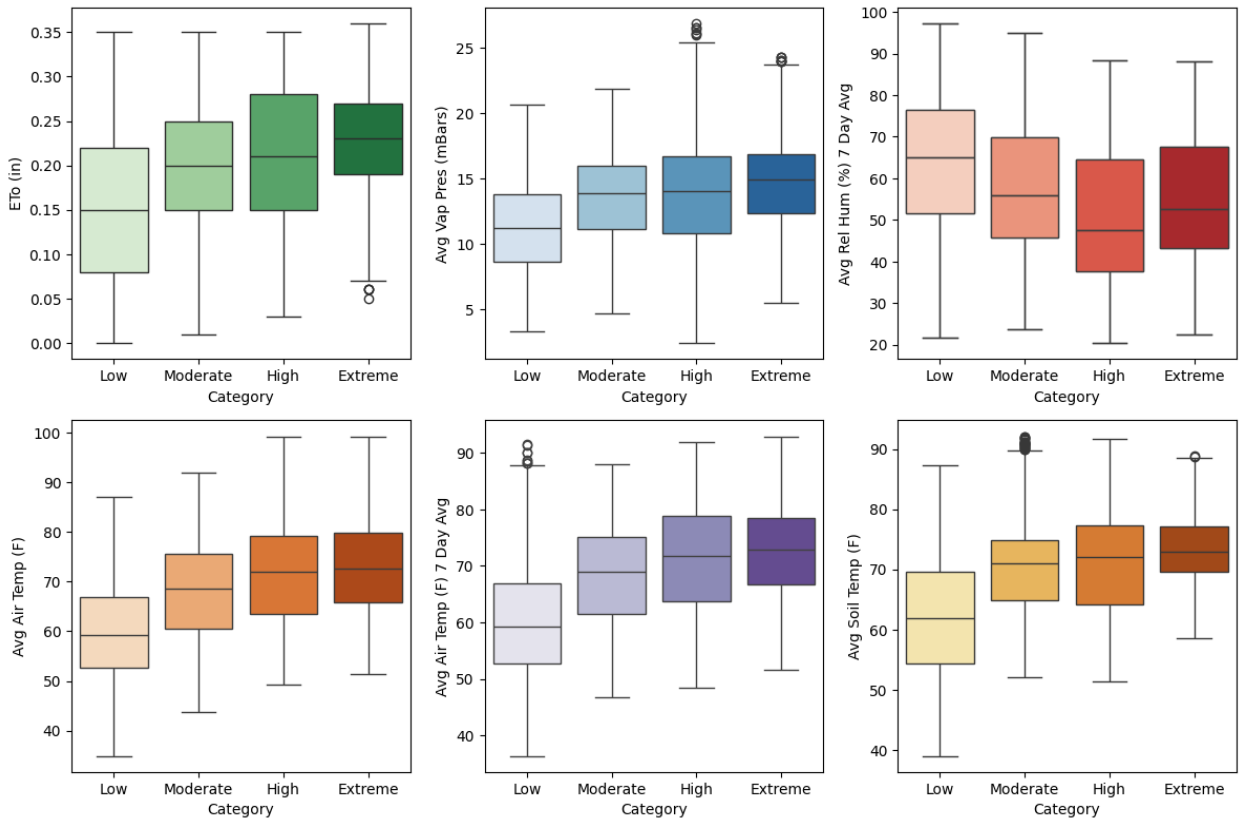
Engineered Features

After examining the interaction plots of our initial proposed engineered combination features, it was determined that the specific combinations we have chosen failed to add consistent correlations to the project and we decided to drop them from the model.



Included Features

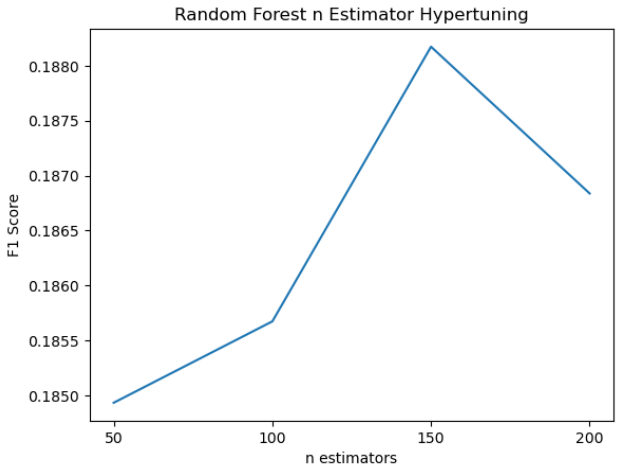
All of the variables chosen to model demonstrate consistent correlation with fire severity with the only exception being the average relative humidity 7-day average. Although the variable boxplot hints at a more complex relationship, we included it to represent the moisture supply.

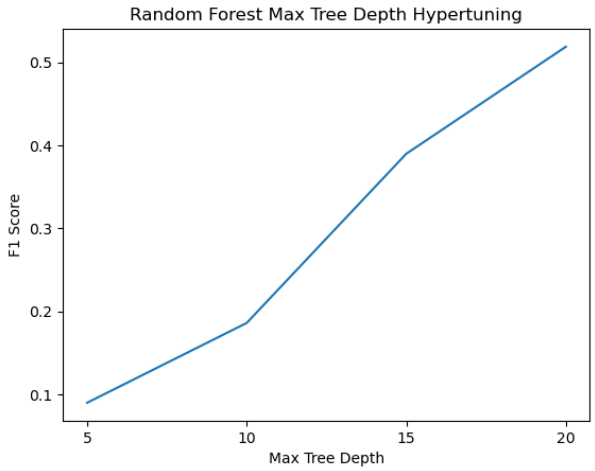


Modeling

Several tuning steps were performed to construct the best model to predict fire severity. Each model underwent hypertuning testing with variations in key individual parameters and each were also tested with different sample balances. The majority class, days that exhibited no fire (or Low) severity, outnumbered all the other minority classes by a factor of 20x. This class imbalance was addressed in three ways, by utilizing in-model methods, undersampling of the majority class, and oversampling of the minority classes. The undersampling exhibited consistent poor performance. Oversampling offered little to no improvement with massive increases in model compute time. The in-model methods offered the best performance with moderate compute time, therefore, hypertuning was conducted without artificial sampling.

Random Forest Model Performance





Optimum Random Forest Build

```
rf = RandomForestClassifier(    n_estimators=150,
                                max_depth=20,
                                min_samples_split=10,
                                max_features = 'sqrt',
                                class_weight='balanced')
```

Best Random Forest Performance With built in class balancing

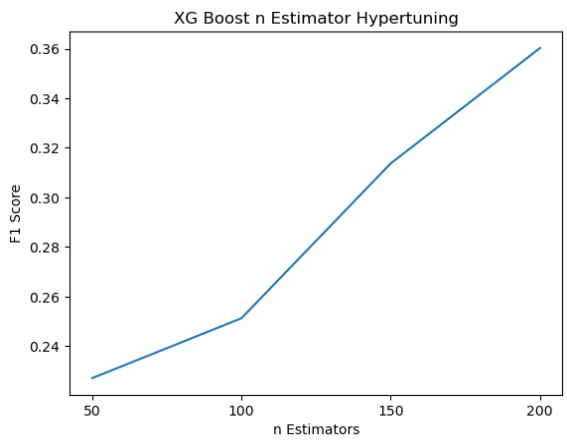
Random Forest K Fold Training			
Category	**Precision**	**Recall**	**F1-Score**
Extreme	0.58	0.5	0.54
High	0.6	0.32	0.42
Moderate	0.53	0.32	0.4
Low	0.97	0.98	0.97

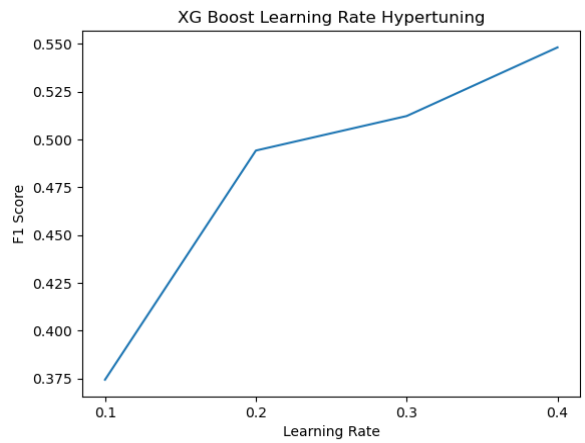
Random Forest Test Set			
Category	**Precision**	**Recall**	**F1-Score**
Extreme	0.55	0.55	0.55
High	0.61	0.35	0.44
Moderate	0.54	0.34	0.42
Low	0.97	0.98	0.98
Macro Avg	**0.67**	**0.56**	**0.6**

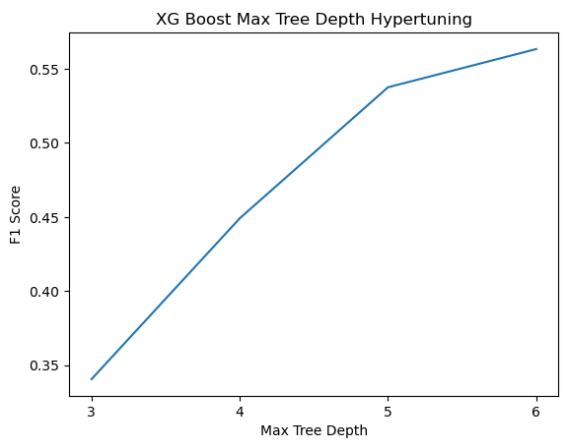
Before hypertuning, this model performed poorly due to likely underfitting the data. The F1-score averaged 10 to 20 percent for the minority cases and an 80 percent score for the majority. Expanding the max depth improved performance dramatically allowing the model to capture more complex relationships among the weather factors.

Experimentation with increased maximum tree depth may improve performance but with the cost of lengthier compute times.

XG Boost







Optimum XG Build

```
xgb = xgb.XGBClassifier(
    objective='multi:softmax',
    num_class=4,
    learning_rate=0.4,
    max_depth=6,
    n_estimators=200
)
```

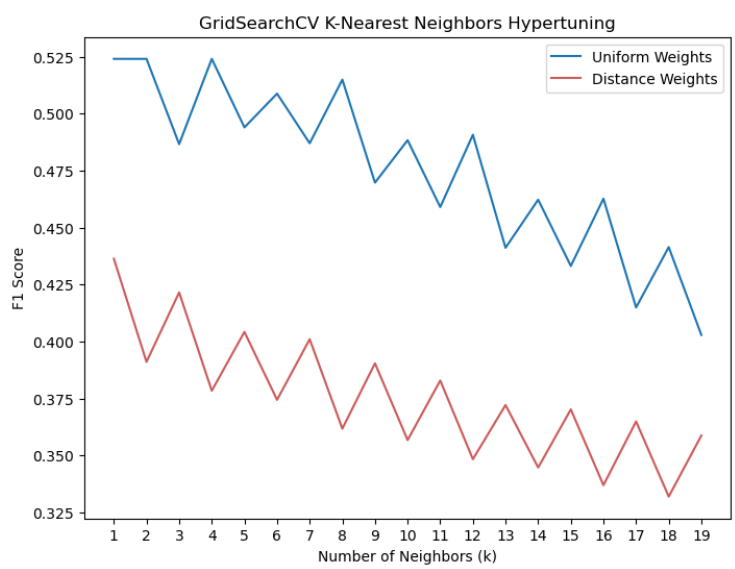
Best XGB Performance

XG Boost K Fold Training			
Category	Precision	Recall	F1-Score
Extreme	0.74	0.44	0.55
High	0.73	0.29	0.41
Moderate	0.71	0.29	0.41
Low	0.96	0.99	0.98

XG Boost Test Set			
Category	Precision	Recall	F1-Score
Extreme	0.8	0.46	0.58
High	0.78	0.34	0.47
Moderate	0.7	0.28	0.4
Low	0.96	0.99	0.98
Macro Avg	**0.81**	**0.52**	**0.61**

Like the random forest model above, this method was underfitting the data before hypertuning. Hypertuning graphs indicate that performance may be boosted with additional n estimators and a larger max tree depth. XG Boost exhibits modest F1 scores while also producing a greater precision for all classes.

K Nearest Neighbors (KNN)



Optimum KNN Build

```
knn = KNeighborsClassifier(n_neighbors = 4, weights = 'uniform')
```

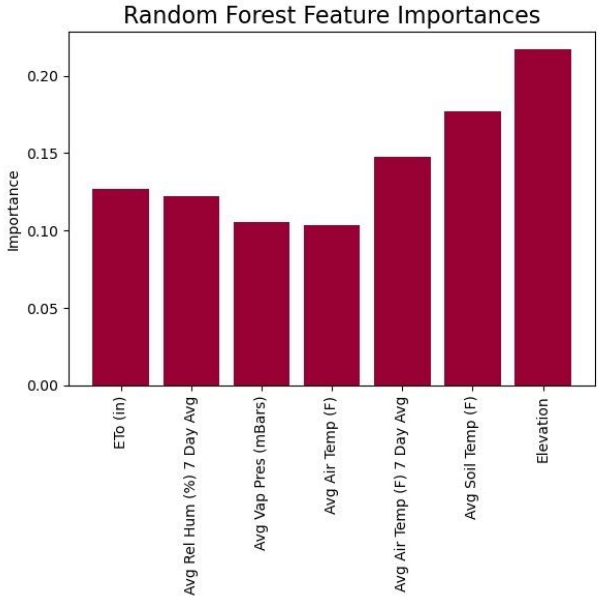
Best KNN Performance

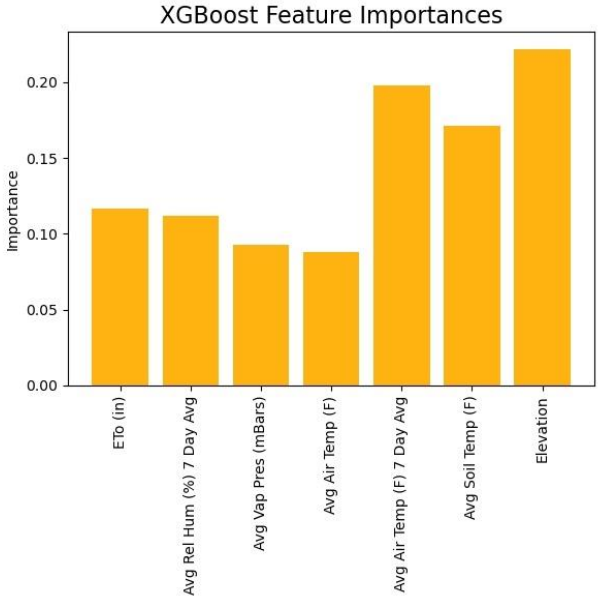
KNN K Fold Training			
Category	Precision	Recall	F1-Score
Extreme	0.46	0.37	0.41
High	0.46	0.26	0.33
Moderate	0.56	0.15	0.23
Low	0.96	0.99	0.97

KNN Test Set			
Category	Precision	Recall	F1-Score
Extreme	0.48	0.39	0.43
High	0.48	0.32	0.38
Moderate	0.52	0.14	0.23
Low	0.96	0.99	0.97
Macro Avg	**0.61**	**0.46**	**0.5**

K Nearest Neighbors performed the poorest among the models indicating that the method may not be a good fit for the data. KNN is more sensitive to higher dimensionality and class imbalances, both of which are present in the weather data.

Feature Contributions





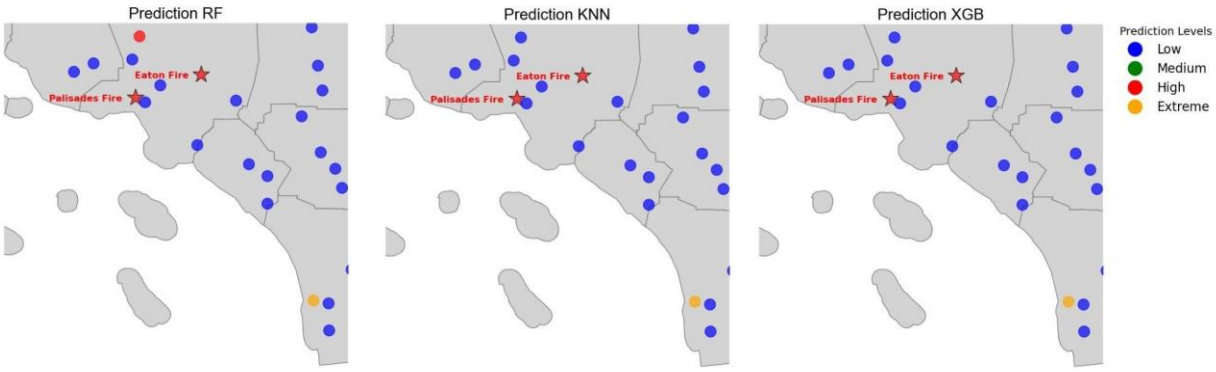
The bar plot of feature importances reveals that elevation was the most important feature in the prediction. This variable was included for research question 2 which examines the geographical effect on the models. Of the weather factors, average soil temperature, an indicator of the dryness of the soil and vegetation, and average air temperature 7 day average, a drought indicator, contribute the most to the predictions.

Results

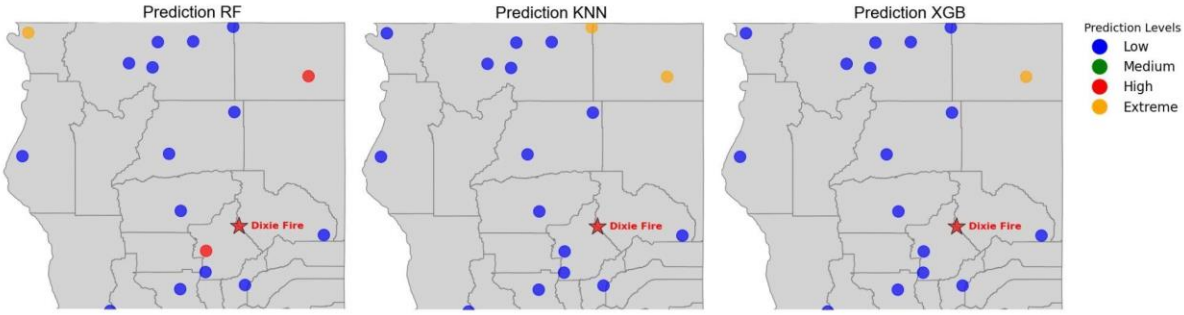
All the models have poor recall scores and struggled the most with differentiation between the middle two classes. Factors besides weather most likely play the strongest role in separating these classes. Overall, the optimal XG boost, and random forest models performed moderately well while the KNN model struggled to make accurate predictions. This suggests that a tree-based approach is best due to the fact that the data is highly imbalanced and that there may be more complicated relationships between the variables. Further exploration of these relationships with combinations or transformations may yield better results.

Predictions were made with CIMIS weather data collected for January of 2025 for the Southern California region and for July of 2021 for Northern California. The devastating Palisades and Eaton fires were ignited on January 7th, 2025, and proceeded to spread rapidly throughout the week. The Dixie fire ignited on July 13, 2021, and destroyed over a thousand structures and caused one fatality.

Southern California Predictions 01-07-2025



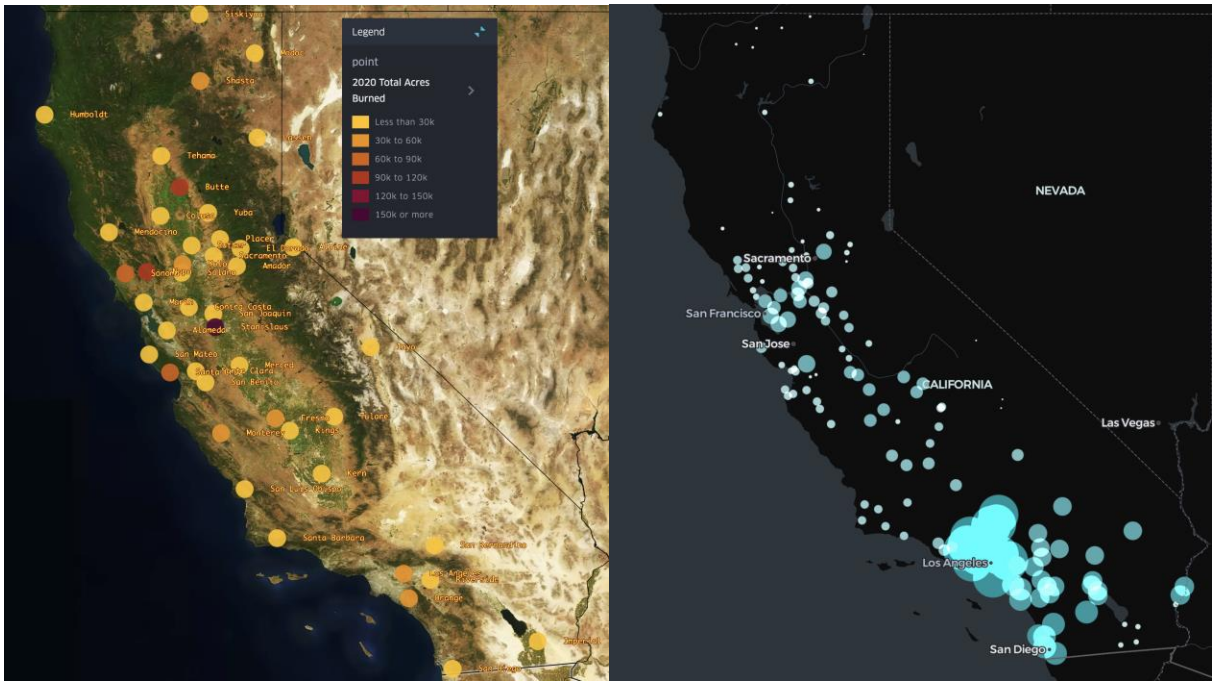
Northern California Predictions 07-13-2021



Although the models did not make an extreme prediction at the direct site of the fires, some did have predictions in the surrounding region. However, these models make strides towards demonstrating the relationship between weather conditions and the severity of a fire. Additional positive cases may go a long way in improving predictive capacity. Supplemental data such as information about vegetation in each county (fuel sources), fire history data, and geographical data, may also improve performance. Increasing granularity from the county level down to a more local level may also help.

Research Question 2: How does the population in urban areas impact correlated wildfire weather factors?

On the lower left, the map of total acres lost due to wildfire in 2020 was generated using Kepler open-source tools. When comparing this map to the density of the population map on the right, counties with higher population density, which are urban areas, seem to be affected a little less than counties with lower population density.



To see if population impacts the weather conditions contributing to the wildfires' severity, we used the same machine learning approaches as we used in research question 1 for this question. The dataset was segmented into three subset of data urban, semi-urban, and rural. We used the information from the California State Association of Counties website to determine different areas of California counties. Urban areas have the highest population density, followed by Semi-urban areas with

lower population density, and Rural areas with the lowest density. California's counties are classified as listed below

- Urban: Alameda, Contra Costa, Fresno, Los Angeles, Orange, Riverside, Sacramento, San Bernardino, San Diego, San Francisco, San Joaquin, San Mateo, Santa Clara, Ventura.
- Semi-urban: Butte, Imperial, Kern, Marin, Merced, Monterey, Napa, Placer, San Luis Obispo, Santa Barbara, Santa Cruz, Shasta, Solano, Sonoma, Stanislaus, Tulare, Yolo.
- Rural: Alpine, Amador, Calaveras, Colusa, Del Norte, El Dorado, Glenn, Humboldt, Inyo, Kings, Lake, Lassen, Madera, Mariposa, Mendocino, Modoc, Mono, Nevada, Plumas, San Benito, Sierra, Siskiyou, Sutter, Tehama, Trinity, Tuolumne, Yuba.

To make it easier to distinguish among the three areas when looking at the confusion matrix, we assigned each area into a different color: urban is blue, rural is green, and semi is orange. Each level of fire severity is also encoded: Extreme = 0, High = 1, Low = 2, Moderate = 3.

For evaluation metrics, we focused on the F-1 score as it is the balanced combination of precision and recall. Precision is the measurement of true positives among positive predictions made by the model, and recall is the measurement of true positive prediction among all actual positives. F-1 score is calculated as below:

$$\text{F-1} = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Since it is a good balance between recall and precision, the F-1 score is useful to use in evaluating the uneven distribution of classes in our dataset, as fires identified as Low class have higher instances than Extreme, Moderate, and High classes. In addition, we focused more on the Extreme class as fires in this class have more negative impact than other classes.

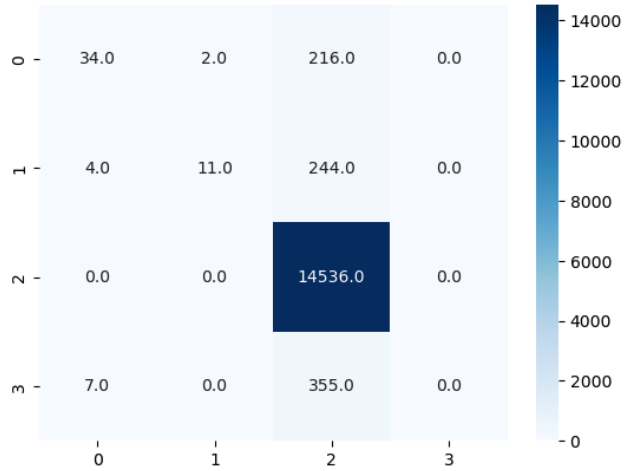
Model Results and Evaluation Metrics Analysis

Urban Area

- Random Forest method

Random Forest:

	precision	recall	f1-score	support
Extreme	0.76	0.13	0.23	252
High	0.85	0.04	0.08	259
Low	0.95	1.00	0.97	14536
Moderate	0.00	0.00	0.00	362
accuracy			0.95	15409
macro avg	0.64	0.29	0.32	15409
weighted avg	0.92	0.95	0.92	15409



As observed above, all the scores of the model on Extreme, Moderate, and High classes are very low compared to the score of the Low class. For instance, Low class has a precision score of 0.95, a recall score of 1.00 (which is too good to be true), and F-1 score of 0.97. Looking more closely at the precision and recall scores of Extreme classes, we recognized a big difference: 0.76 for precision score and only 0.13 for recall score. Below are the precision and recall formulas:

$$\text{Precision} = \frac{True\ Positive}{True\ Positive + False\ Positive}$$

$$\text{Recall} = \frac{True\ Positive}{True\ Positive + False\ Negative}$$

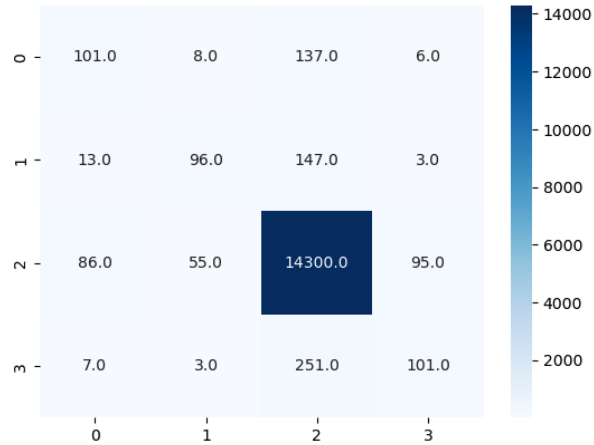
Looking at the confusion matrix generated by this model, for the Extreme class, we have True Positive = 34, False Positive = 4 + 7 = 11, and False Negative = 2 + 216 = 218. The high precision score implies that out of 45 fires predicted as Extreme class, only 34 of them belong to Extreme class. The low recall score implies that out of 252 actual fires classified as Extreme class, only 34 of them are predicted as the corrected class. The similar scenario happens to High class, where the precision score is very high, but the recall score is extremely low. Therefore, the F-1 score for Extreme class is 0.23. Predicting Extreme class fires as other classes could have a negative impact on the affected area as the authorities would not be able to react in time. Next, we proceeded to use the other methods to see if there is a significant difference in the performance.

- KNN method

```
KNN:
              precision    recall  f1-score   support

     Extreme       0.49      0.40      0.44       252
        High       0.59      0.37      0.46       259
         Low       0.96      0.98      0.97     14536
    Moderate       0.49      0.28      0.36       362

    accuracy                           0.95     15409
   macro avg       0.63      0.51      0.56     15409
weighted avg       0.94      0.95      0.94     15409
```



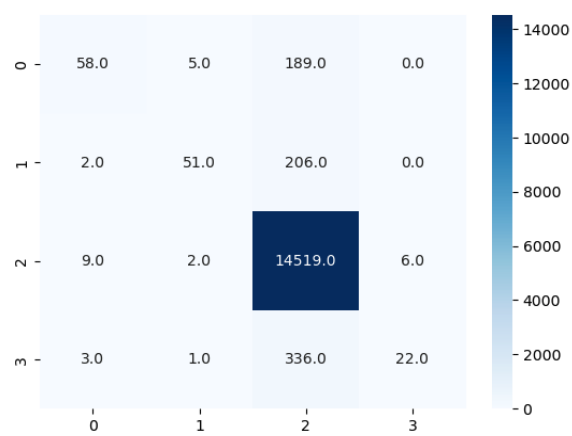
In KNN method, the F-1 score for predicting Extreme risk (along with High and Moderate risk levels) are still lower than F-1 score for predicting Low risk. However, it is higher than the F-1 score in the previous model (0.44 in this model compared to 0.23 in the previous one). In addition, the precision and recall scores do not have a big gap as in the Random Forest model. From this observation, we see that KNN model did better than Random Forest model. We proceeded to use the third method, Gradient Boosting to see if the performance improved.

- Gradient Boosting method with learning_rate=0.1

```
Gradient Boosting:
              precision    recall  f1-score   support

     Extreme       0.49      0.40      0.44       252
        High       0.59      0.37      0.46       259
         Low       0.96      0.98      0.97     14536
    Moderate       0.49      0.28      0.36       362

    accuracy                           0.95     15409
   macro avg       0.63      0.51      0.56     15409
weighted avg       0.94      0.95      0.94     15409
```



In this model, the performance metrics are very similar to the KNN model: average macro score for F-1 is 0.56, and F-1 score for predicting Extreme class is 0.44.

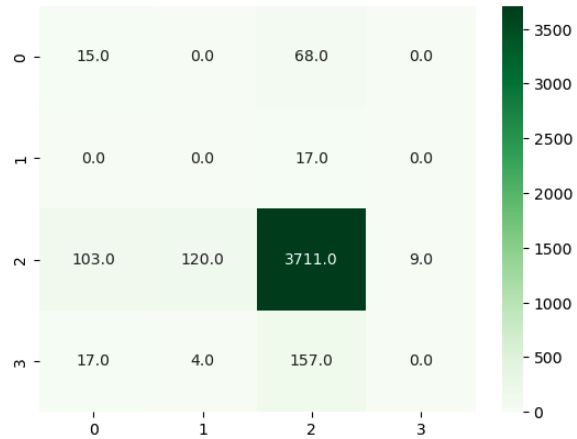
Next, we applied the same methods to the other subsets of data (Rural and Semi-urban).

Rural Area

- Random Forest, with max_depth = 3 and n_estimators = 200

Random Forest:

	precision	recall	f1-score	support
Extreme	0.11	0.18	0.14	83
High	0.00	0.00	0.00	17
Low	0.94	0.94	0.94	3943
Moderate	0.00	0.00	0.00	178
accuracy			0.88	4221
macro avg	0.26	0.28	0.27	4221
weighted avg	0.88	0.88	0.88	4221



- KNN, with k = 3

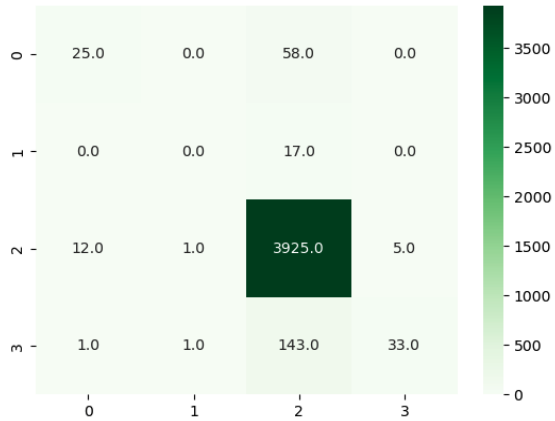
KNN:

	precision	recall	f1-score	support
Extreme	0.60	0.60	0.60	83
High	0.33	0.18	0.23	17
Low	0.97	0.98	0.98	3943
Moderate	0.67	0.56	0.61	178
accuracy			0.95	4221
macro avg	0.64	0.58	0.60	4221
weighted avg	0.95	0.95	0.95	4221

- Gradient Boosting, with learning rate = 0.1

Gradient Boosting:

	precision	recall	f1-score	support
Extreme	0.66	0.30	0.41	83
High	0.00	0.00	0.00	17
Low	0.95	1.00	0.97	3943
Moderate	0.87	0.19	0.31	178
accuracy			0.94	4221
macro avg	0.62	0.37	0.42	4221
weighted avg	0.93	0.94	0.93	4221

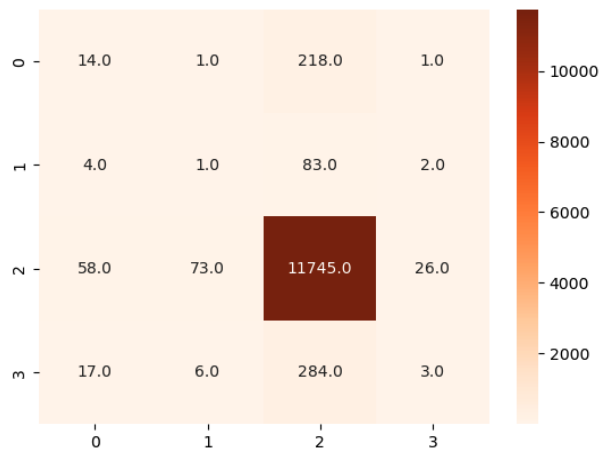


Semi Area

- Random Forest, with n_estimators = 200

Random Forest:

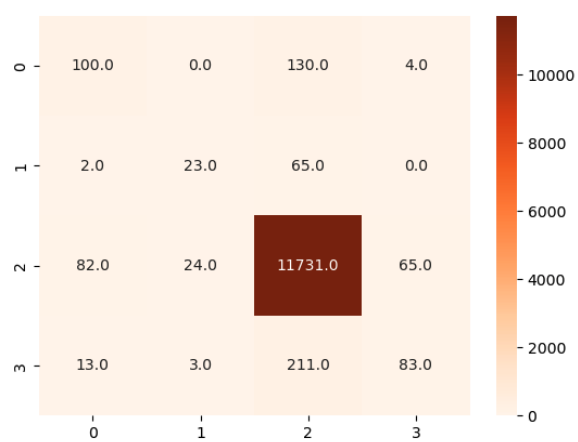
	precision	recall	f1-score	support
Extreme	0.15	0.06	0.09	234
High	0.01	0.01	0.01	90
Low	0.95	0.99	0.97	11902
Moderate	0.09	0.01	0.02	310
accuracy			0.94	12536
macro avg	0.30	0.27	0.27	12536
weighted avg	0.91	0.94	0.92	12536



- KNN, with k = 3

KNN:

	precision	recall	f1-score	support
Extreme	0.51	0.43	0.46	234
High	0.46	0.26	0.33	90
Low	0.97	0.99	0.98	11902
Moderate	0.55	0.27	0.36	310
accuracy			0.95	12536
macro avg	0.62	0.48	0.53	12536
weighted avg	0.94	0.95	0.95	12536

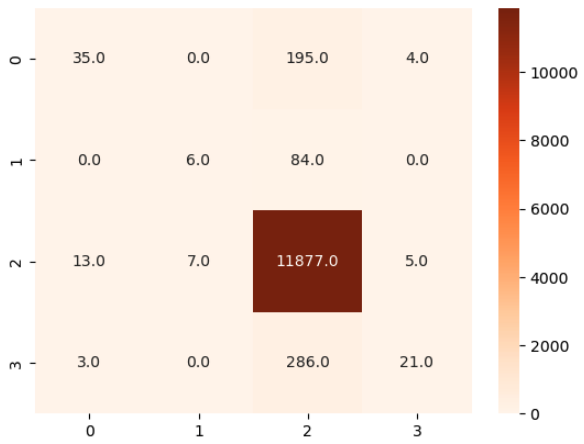


- Gradient Boosting, with learning rate = 0.1

```
Gradient Boosting:
              precision    recall  f1-score   support

     Extreme       0.69      0.15      0.25       234
        High       0.46      0.07      0.12        90
         Low       0.95      1.00      0.98     11902
    Moderate       0.70      0.07      0.12       310

    accuracy                           0.95     12536
   macro avg       0.70      0.32      0.37     12536
weighted avg       0.94      0.95      0.93     12536
```



Results:

Through the results of the models, we noticed that the precision scores are always higher than recall scores in all models, and this difference happened not only in Extreme class, but also in High and Moderate classes. The precision scores in the other subsets of data are also higher than recall scores, and this difference applies to Extreme, High and Moderate classes. High precision and low recall suggest that the models are more precise in identifying true positives in their positive prediction but fail to capture the substantial number of actual positives. Since the difference between those two scores is significant in some models, and both scores are not very high (as in Low class), low F-1 scores happened as a result for all models across all subsets of data.

In addition, we observed KNN and Gradient Boosting seem to predict the fire risk classes better than the Random Forest model. In the Rural dataset, KNN is the model performing the best on predicting Extreme fire risk compared to Random Forest and Gradient Boosting, with F-1 score and macro average F-1 score of 0.60. Similar results

in Semi-urban dataset, where KNN is the best model with F-1 score for predicting Extreme class is 0.46, and macro average F-1 score is 0.53.

Applying k-fold cross validation and hypertuning:

After initially generating the models to predict the fire severity for each class (urban, semi-urban, and rural), we performed k-cross validation on each model to minimize the risk of overfitting with the number of folds is three. We used macro for F-1 and recall scores since we wanted to address the importance of all fire risk levels equally regardless of the occurrence of each class.

Below is the table of average scores (F-1 and Recall) after applying k-fold cross validation for each model, with highlighted values being the highest score when comparing among different models of the same class (urban, semi-urban, or rural).

	Random Forest	KNN	Gradient Boosting
Urban	F-1 = 0.3330 Recall = 0.3050	F-1 = 0.5145 Recall = 0.4705	F-1 = 0.4283 Recall = 0.3674
Semi-Urban	F-1 = 0.2478 Recall = 0.2522	F-1 = 0.5147 Recall = 0.4751	F-1 = 0.3884 Recall = 0.3359
Rural	F-1 = 0.2895 Recall = 0.2762	F-1 = 0.5587 Recall = 0.5191	F-1 = 0.4693 Recall = 0.3984

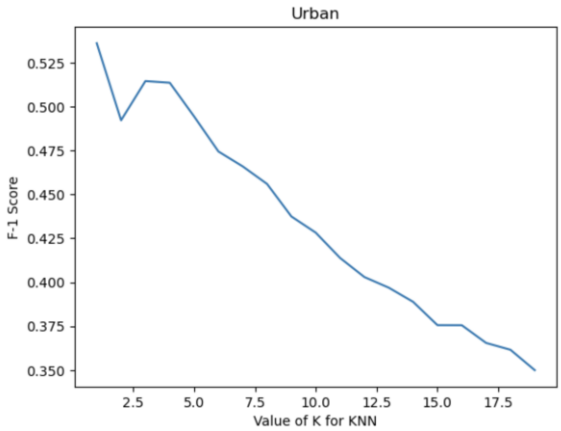
Through this comparison, **KNN** seems to perform the best across all classes with k=3 before hypertuning.

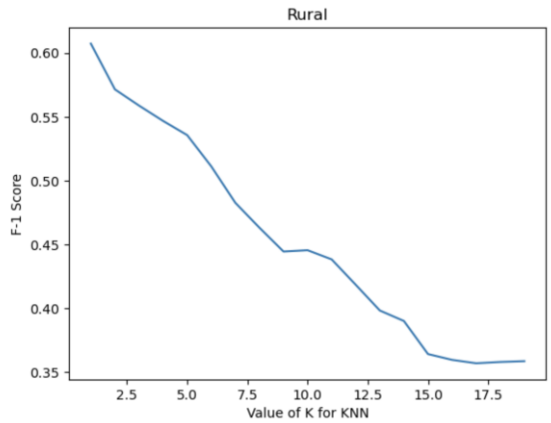
We proceeded to apply hypertuning using GridSearch, with a cross-validation generator set to 3. The purpose is to find the parameters that could produce a better

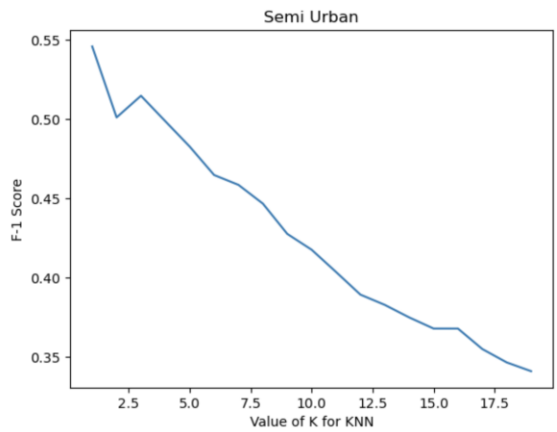
performance The parameters used in hypertuning are different values of tree depth (5,7,10) and numbers of estimators (300, 500, 700) for Random Forest, the k-value for KNN in a range from 1 to 20, and the learning rate for XGBoost (0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9). The scoring metrics are set with f1_macro and recall_macro.

	Random Forest	KNN	Gradient Boosting
Urban	F-1 = 0.4102 Recall = 0.3557	F-1 = 0.5360 Recall = 0.5337	F-1 = 0.4822 Recall = 0.4107
Semi-Urban	F-1 = 0.3390 Recall = 0.3008	F-1 = 0.5458 Recall = 0.5493	F-1 = 0.4946 Recall = 0.4426
Rural	F-1 = 0.3895 Recall = 0.3951	F-1 = 0.6073 Recall = 0.6288	F-1 = 0.4928 Recall = 0.4857

From the results above, the best model predicting fire risk before and after hypertuning is KNN, with k=1. To have a better vision of how k = 1 resulted in the best F-1 scores, the plots below show how F-1 score changes when applying different k that ranges from 1 to 20. As shown in the plots, F-1 scores drop drastically as k increases.







After having value 1 as the number of k in KNN, we ran the models again with n_neighbors = 1.

Urban, KNN model:

	precision	recall	f1-score	support
Extreme	0.47	0.49	0.48	252
High	0.48	0.46	0.47	259
Low	0.97	0.97	0.97	14536
Moderate	0.36	0.37	0.37	362
accuracy			0.94	15409
macro avg	0.57	0.57	0.57	15409
weighted avg	0.94	0.94	0.94	15409

Rural, KNN model:

	precision	recall	f1-score	support
Extreme	0.59	0.61	0.60	83
High	0.29	0.24	0.26	17
Low	0.97	0.97	0.97	3943
Moderate	0.58	0.58	0.58	178
accuracy			0.94	4221
macro avg	0.61	0.60	0.60	4221
weighted avg	0.94	0.94	0.94	4221

Semi-urban, KNN model:

	precision	recall	f1-score	support
Extreme	0.50	0.54	0.52	234
High	0.35	0.38	0.36	90
Low	0.97	0.97	0.97	11902
Moderate	0.38	0.42	0.40	310
accuracy			0.94	12536
macro avg	0.55	0.58	0.56	12536
weighted avg	0.94	0.94	0.94	12536

F-1 scores for predicting the Extreme class of Urban, Rural, and Semi-Urban are 0.48, 0.60, and 0.52 respectively, and macro average F-1 scores are 0.57, 0.60, and 0.56 respectively. There is no significant difference among macro average F-1 scores of all three areas since the difference is only 0.03 between Rural and Urban, and 0.01 between Semi-urban and Urban. The difference of F-1 scores of Extreme class between Urban and Rural (0.12) shows a bigger difference, although not too significant.

Conclusion:

Even though all the evaluation metrics showed that the weather conditions are not the best factors to predict the fire risk, our goal in this task is not to see if how well the weather conditions could predict the fire risk, but to see if there is a significant difference of prediction ability among three areas (Urban, Rural, and Semi-urban). From all the tests and the evaluation metrics, we determined that the population does not impact the weather conditions contributing to wildfire severity due to the insignificant differences in fire risk prediction ability among all three areas. All the evaluation metrics for all the models have a F-1 and recall scores lower than 0.7, and the only class with exceptional high scores is Low class (larger than 0.9). Even when segmenting the dataset into three subsets associated with three areas, the performance is not notably different. Based on this observation, it is very likely that the area and its population does not have an impact on the weather conditions contributing to the wildfire risk. Other factors that could contribute significantly to the wildfire risk and impact the weather conditions could be topography and the variety of vegetation.

Ethical Recommendations

With the increasing frequency and intensity of wildfires around the globe, communities, businesses, and governments are more motivated than ever to take action to predict, detect, and prevent wildfires. The ability to limit and or prevent the devastating impact wildfires have on communities such as damaging housing and critical infrastructure, hindering the transportation of necessities like food and water, and the potential for loss of life, would be life-altering. However, not all parties may be interested in predictive technology for the same reasons. While the technology to identify communities and regions most at risk of wildfires should be used in an ethical and equitable manner, some may weaponize the data and use it with ill intentions.

Importance of Data

It is important to first note the dataset itself can also pose ethical issues due to both conscious and unconscious biases. The method of data collection may be biased due to regions or demographics that were included or excluded from the set. The accuracy of our predictions depends on the quality and completeness of the dataset. To bolster the reliability and completeness of our dataset, multiple datasets have been sourced and collected by federal agencies before blending them to ensure more equitable and complete testing and predictions. Sources in our blended dataset also include environmental monitoring systems, fire stations, wildfire damages, and fire history data. As our team sifted through the data it became apparent that rural and sparsely populated regions had a lower representation in the statistics. This may be due

to reports of wildfires depending on human accounts and recording, leading to areas with higher populations reporting more often. Due to these, urban areas have the potential to be overemphasized in models. Older regions can also be affected by insufficient or incomplete research and outdated data, leading to overreliance on subpar data. These regions face other implications due to this underrepresentation, such as less fire-preparedness compared to higher-density urban areas with more resources and may receive less or inadequate attention from emergency response services. Using historical data, the focus will be servicing more established regions and providing insights for these areas which have the most complete datasets.

Good vs Bad Use Cases

There is a risk of businesses misusing these predictive models to their benefit and the detriment of communities. According to Picchi, in July of 2024, State Farm dropped approximately 1,600 fire coverage policies for the residents of Palisades months before the devastating Palisades Fires swept through. It was also found that State Farm dropped over 2,000 policies in two other Los Angeles ZIP codes as well, including Brentwood, Calabasas, Hidden Hills, and Monte Nido (Picchi, 2025). The sudden cancellation of insurance policies left many homeowners either without coverage or paying higher rates for even less coverage (Picchi, 2025). Insurance companies have the potential and the incentive to utilize fire prediction models to target at-risk communities with higher pricing, offering dwindling coverage, or outright canceling their policies leaving them with no coverage despite paying their dues. It is up to local communities and government agencies to utilize these models to benefit

everyone while limiting loss of life and structural damage. The ethical use of fire prediction models will benefit society by accurately anticipating their occurrence and properly allocating necessary firefighting and recovery resources for regions based on risk. Agencies such as the Office of the State Fire Marshal in California already have fire prevention and land use planning programs to protect and educate the public and could further improve their efforts with the use of accurate prediction models as technology continues to advance. Institutions such as UC Davis are contributing to wildfire monitoring as well through the use of drones that are capable of measuring air quality, temperature, humidity, and wind speeds among other things. Their research has produced drones that can detect conditions related to high-fire risk and sweep areas for indications of a fire at a faster rate and larger scale than humans are capable of (Anderson, 2023). Pairing fire-prediction models with UC Davis' drone research could improve data collection for future prediction models and limit damage with faster fire detection. With the ethical applications of these technologies in fire-risk communities, many individuals, families, and businesses will benefit from a better allocation of emergency resources and ideally, will not have to suffer heavy losses to wildfires.

Earning Public Trust with Tools

While fire prediction models can be useful tools, it must be emphasized that they must be used as supplementation to other traditional methods. We argue strongly against emergency services solely relying on these forecasts but instead encourage them to be used as proactive tools in conjunction with expert opinions and resources. It is imperative that they are not the only tool used as it is well known that models such as

XGBoost are a type of black box model, meaning its decision-making is almost impossible to comprehend due to program encapsulation. Overreliance may stir distrust in the general public due to concerns about accountability and the consequences of AI resource allocation, specifically if an inappropriate decision is made in an emergency scenario. For example, if the model generates a false negative for fire risk, evacuations may be initiated too late, resulting in preventable property destruction, loss of life, and loss of public trust. Additional techniques we employed to improve model performance include feature importance analysis and Shapley Additive Explanations (SHAP), which greatly clarify the environmental parameters that are used to classify wildfire risk. These approaches allow us to determine which factors have the highest predictive power, allowing researchers and even emergency responders to better comprehend the logic behind AI's action. Proper application of our models combined with users having a better understanding of the AI's logic will lead to improved resource allocation and ultimately, higher quality of life for many communities.

Challenges

A primary issue for our team was the difficulty of choosing a suitable dataset to analyze. At the beginning of the project, the team primarily focused a lot of effort on trying to locate datasets that would correspond with the research questions. Unfortunately, most of the internet datasets that were present lacked a certain level of detail necessary for analysis. Some datasets exhibited a vague, as in, we captured multifaceted aspects of wildfires without pinpointing specifics that could be useful for predictions during certain fire scenarios. Some also lacked certain values or had

discrepancies, rendering all of the datasets nearly impossible to work with without some degree of modification. There was also the additional problem of scope regarding the environmental factors necessary for wildfire behavior prediction, such as temperature, humidity, soil moisture, wind speed, and more. The challenge of collecting a workable dataset led to drastic changes in the team's project objectives, limiting the range of variables that could be analyzed, and as a result, needed to be reevaluated concerning the existing information. Due to this step in narrowing the scope, the project faced some delays since various rounds of dataset sampling needed to be done before settling on a feasible option. When the appropriate dataset was found, the team was met with the challenge of integrating and cleaning the data. The sources of the dataset included satellite images, weather observations collected from various sensors, and records of fires. These sources posed significant problems in attempting to combine them into one cohesive dataset because all the datasets had different formats, resolution, and update rates. For instance, sensor data had much higher spatial resolution than satellite imagery. This resulted in inconsistencies in the geographical precision of the final dataset. In many instances, the team was forced to make changes to data formats, input missing information, and delete incorrect information, which proved to be extremely time consuming.

Additionally, the out-of-date some of the data proved to be problematic as well since the team had to make sure data sources double checked with the correct information. Even after cleaning and integrating the data, the team faced much difficulties in analyzing the cleaned data correctly. Given the many variables, each one acting differently on wildfire behavior, determining which features to use in fire risk

predictions was problematic. Factors such as temperature anomalies, wind, and soil moisture, and how those all interrelate requires an inordinate amount of expertise analysis. Regardless of the chosen crucial factors, the team engaged other challenges in modeling wildfire behavior as it is complex and not always linear. Certain model outputs were erratic, such that some predictions would not be consistent across different fire events or regions, which made it impossible to produce dependable outcomes.

Constructing and verifying the predictive models was another difficult task. Random Forests was chosen as the best alternatives toward predicting wildfire occurrences, while the team tried to accomplish the task with the help of machine learning. Nevertheless, the models Wildfires are characterized by a broad range of variability, so the outcomes are highly capricious. Along with the environmental factors such as wind and humidity, which tend to remain non-linear and interact in rather erratic manners, the models also struggled with overfitting, because we did well on the training data but poorly on the actual new data. The team adjusted the parameters, modified the model features, and even changed the algorithms to address the performance issues which were prominent in every iteration of model training. Out of the many algorithms the curation of a rich dataset, selecting the right features, which would be the most informative about fire behavior, was the most challenging as it contains a lot of noise from the environment. And, while the models are rich in detail, the older methods still leave much to be desired with regard to the fire dynamics that could render effective predictive models highly inaccurate. Wildfires are affected by numerous external factors, some of which are almost impossible to gauge or anticipate, making it next to

impossible for any model to generate accurate outcomes consistently. While the team attempted to calibrate the models, the model performance kept changing, compelling the team to change their assumptions again and again. These problems were compounded by the differences in wildfire patterns in different parts of California. A model that performed accurately in one area would likely not be valid for another area because of differences in its vegetation, topography, or climate. This required splitting the work into smaller models which was more difficult for the project and increased the workload.

At the final stage, our team worked hard on translating our results into data visualizations after refining the models because it was the most challenging part of our work. As responding to emergencies is very complex, creating visualizations that provide actionable insights proved to be difficult because of the advanced nature of the data. The team also sought to develop and design user friendly dashboards and maps that were sophisticated enough to accurately portray the intended meaning. Simultaneously, the algorithms needed to process and analyze data in real time and provide insights without overwhelming users. Furthermore, the visualizations had to be intelligible not only by the users but also by various experts in data analytics. Striking the right balance of depth and sophistication of information with ease of comprehension for non-expert users proved to be one of the greatest problems of the project since the target audience, which was the emergency managers and responders, needed to benefit from the final product.

Recommendations and Next Steps

Recommendations

As previously mentioned, we strongly encourage ethical and equitable use of our research to provide the maximum benefit for those who are most at risk of wildfires in these communities. Due to the intricate nature of our research questions we struggled to identify any strong correlations between weather factors or populations in relation to wildfires and their severity. Through the creation of data visualizations, it became apparent that we were lacking enough data to come to any strong conclusions. Since our research is incomplete, it should not be used in any substantial way by emergency services. Further research should be conducted to identify other factors that may play a role in wildfire severity such as additional topographic information. Specifically, topographical data regarding vegetation varieties and densities. Organizations such as California's Chaparral Institute emphasize the importance of native plant life cycles and how they are linked to and encourage naturally occurring wildfires in the biome. Including this information in future analyses would be highly beneficial since a relationship between vegetation and wildfires has already been established by other researchers. It is likely that a relationship between population and wildfire intensity may become apparent with the additional supplemental data, leading to another opportunity to improve community wildfire preparedness and response with a more reliable prediction model.

Next Steps

The next phases of this capstone project should focus on improving the data processing pipelines and enhancing the predictive models. Incorporating real-time data from NOAA, CAL FIRE, and NASA FIRMS should be automated to ensure that the models are constantly up to date, and more importantly, improving autonomously. Geospatial interpolation can help deal with missing datasets, and automating more processes of wildfire risk assessment will increase job productivity. Also, these newer models should be further optimized by re-training them on newer 2021-2024 wildfire data, utilizing different machine learning and resampling methods for the class imbalance problem such as SMOTE and GAIN-criteria. Another useful step is to try to increase the predictive power beyond the county borders towards the city or district level. This would greatly improve accuracy and thus help in aiding emergency preparedness. The next step would be an effective creation and deployment of an interactive wildfire risk dashboard. Prototype development and real-time deployments along with user authentication systems should be utilized and it is important that the interface is user friendly for both the public as well as emergency responders. Testing of predictive models in real-life emergency scenarios will be effective with the collaboration of CAL FIRE and other local fire departments. The system can be finetuned further through constructive feedback and training workshops with first responders. Research outcomes can be published while collaboration with open sources will ensure advancements in wildfire prediction. In conjunction with government agencies and

private sponsors, funding and policy support will result in the long-term scalability and sustainability of those predictive technologies.

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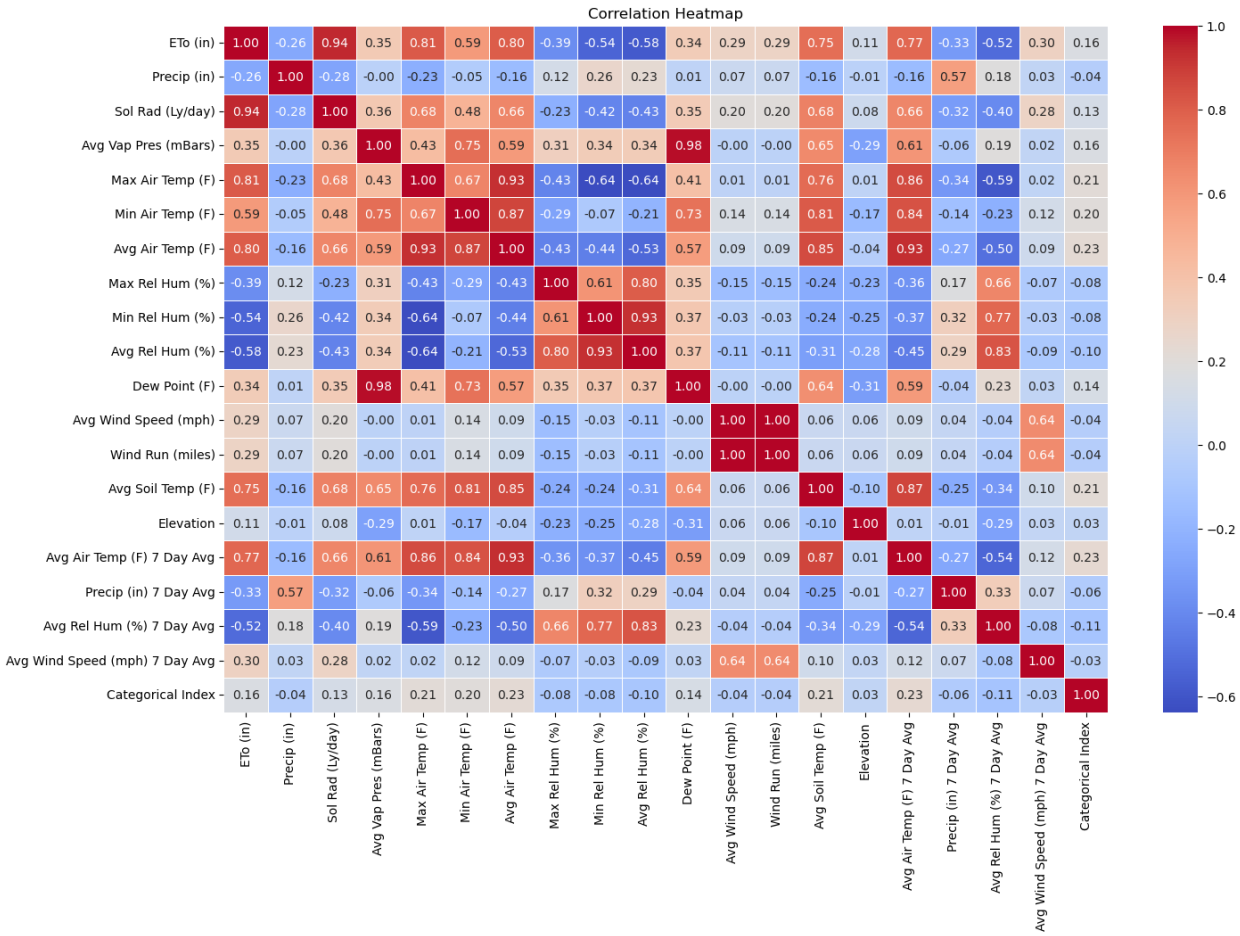
Code

Research Question 1:

https://drive.google.com/file/d/1haH06i9YXLb2YCHLaBnf9HJfFTVtsGQF/view?usp=sharing

Research Question 2:

https://drive.google.com/file/d/12H5VVS34l0p7JJp14isOs0nb122ITfIP/view?usp=drive_link

Appendix

Correlation heatmap of all variables in dataset before variable selection.